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Comparing Human Pose Estimation through deep learning approaches: An overview

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ABSTRACT

In the everyday IoT ecosystem, many devices and systems are interconnected in an intelligent living environment to create a comfortable and efficient living space. In this scenario, approaches based on automatic recognition of actions and events can support fully autonomous digital assistants and personalized services. A pivotal component in this domain is "Human Pose Estimation", which plays a critical role in action recognition for a wide range of applications, including home automation, healthcare, safety, and security. These systems are designed to detect human actions and deliver customized real-time responses and support. Selecting an appropriate technique for Human Pose Estimation is crucial to enhancing these systems for various applications. This choice hinges on the specific environment and can be categorized on the basis of whether the technique is designed for images or videos, single-person or multi-person scenarios, and monocular or multiview inputs. A comprehensive overview of recent research outcomes is essential to showcase the evolution of the research area, along with its underlying principles and varied application domains. Key benchmarks across these techniques are suitable and provide valuable insights into their performance. Hence, the paper summarizes these benchmarks, offering a comparative analysis of the techniques. As research in this field continues to evolve, it is critical for researchers to stay up to date with the latest developments and methodologies to promote further innovations in the field of pose estimation research. Therefore, this comprehensive overview presents a thorough examination of the subject matter, encompassing all pertinent details. Its objective is to equip researchers with the knowledge and resources necessary to investigate the topic and effectively retrieve all relevant information necessary for their investigations.

1. Introduction

Human Pose Estimation (HPE) is a fundamental aspect of computer vision with significant implications for understanding human behavior and interaction in digital environments. Accurate spatial configuration of the human body is essential for various applications such as human–computer interaction, virtual reality, surveillance, and healthcare. With advancing technology, there is an increased demand for advanced pose estimation algorithms to address the challenges associated with capturing and interpreting human motion. This survey paper aims to provide a comprehensive overview of state-of-the-art techniques and methodologies used in HPE by synthesizing diverse research efforts to highlight the evolution of methods along with their underlying principles and applications across different domains. The historical development, from traditional approaches to recent techniques is explored, along with discussions on trade-offs, advantages, limitations, and modalities like RGB imagery and depth sensors employed in pose estimation. Furthermore, the role of deep learning – mainly using convolutional neural networks, recurrent neural networks, and transformers alongside hybrid architectures – which has revolutionized this field, is discuss as well. Furthermore, key benchmarks and datasets are also emphasized throughout this survey, aiming to provide researchers, practitioners, and enthusiasts with an extensive resource for existing knowledge synthesis, while guiding potential future research directions within this dynamic field.

Several survey papers on HPE have been published in recent years. Analysis focusing on deep-learning-based pose estimation methods can be categorized as follows:

- 2D only HPE (Dang et al., 2019; Munea et al., 2020);

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- 3D only HPE (Sarafianos et al., 2016; Wang et al., 2021b);
- both 2D and 3D HPE (Chen et al., 2020; Song et al., 2021b; Zheng et al., 2024; Dubey and Dixit, 2023; Yan et al., 2024).

The main differences between the 2D/3D works compared to ours are that Chen et al. (2020), Yan et al. (2024) specifically examine methods that work with monocular sources, whereas (Song et al., 2021b; Dubey and Dixit, 2023) also focus on other applications besides pose estimation, potentially providing less in-depth HPE analysis, i.e. action recognition and adversary security attacks, respectively. The works by Dubey and Dixit (2023), Yan et al. (2024) are the most recent ones, but also (Zheng et al., 2024) starting with the first version in 2020, and the fifth in early 2023 can be considered quite updated.

Contribution of this work. By unraveling the intricate language of human movements, pose estimation empowers machines to understand, interpret, and respond to human actions. The significance of HPE lies in its ability to capture the essence of human gestures, posture, and gait. We aim to provide helpful insights on the topic to researchers who would like to use HPE in many research fields, including, but not limited to:

- Human-Computer Interaction (HCI): HPE unlocks a new era of intuitive and immersive HCI. By tracking hand gestures and body movements, pose estimation enables users to control virtual interfaces and devices with unprecedented naturalness, e.g. ability to manipulate digital objects, browse through information, or engage with virtual assistants solely through your body language (Tsai et al., 2020). Huo et al. (2023) employed a 3D HPE network for generating animated character motions and tracking robot movements, and developed two systems for human–computer and human–robot interaction (HCI/HRI) applications.
- Action Recognition: HPE is crucial for recognizing human actions, allowing machines to interpret the meaning behind human movements. This capability is used in various fields such as sports analysis, motion capture, surveillance, and robotics by enabling machines to comprehend human actions and react appropriately (Sun et al., 2023). Several works can be found in literature that are using some well-known frameworks for HPE, e.g. OpenPose (Cao et al., 2017), BlazePose (Bazarevsky et al., 2020), to create specific applications for action recognition, like specific yoga poses (Thar et al., 2019; Desai and Mewada, 2023; Upadhyay et al., 2023) or fitness actions (Xu et al., 2023), to count and validate cross fit exercises (Ferreira et al., 2021) and monitor the human body's joints in activities like walking, running and cycling (Pagnon et al., 2021).
- Healthcare: HPE shows great potential in healthcare, especially in rehabilitation and gait analysis. By tracking patient movements and body positions, pose estimation algorithms can help assess injuries, monitor rehabilitation progress, and detect gait abnormalities, aiding in early intervention and improved patient outcomes (Zheng et al., 2024). Regarding the gait analysis, there are applications that are measuring general spatiotemporal and sagittal kinematic gait parameters (Stenum et al., 2021), while others are focusing on measuring the knee flexion and extension angles (Viswakumar et al., 2019) and identifying heel strike and toe-off events timings (Hii et al., 2023). Some examples of successful applications in the rehabilitation field are the measurement of 5 active shoulder ranges of motion (Latreche et al., 2023) and the estimation of the frequencies of 5 repetitive movement tasks (Cornman et al., 2021).
- Retail and Entertainment: HPE has brought about significant changes in the retail and entertainment sectors. Virtual fitting rooms, now enabled by pose estimation, allow customers to virtually test clothing and accessories, offering a more individualized and interactive shopping experience. In gaming, pose estimation is pivotal in motion-capture technology, empowering players to control game characters through their physical movements elevating overall immersion and gameplay quality (Bhardwaj et al., 2024). In animation, Kumarapu and Mukherjee (2021) introduced an innovative and comprehensive framework for frame-by-frame 3D multi-person pose estimation and animation in videos. By leveraging depth information, they accurately localize the 3D relative poses within a 3D environment.
- Social Robotics: HPE is empowering social robots to interact with humans more naturally and effectively. By interpreting human body language, robots can perform deeper conversations, react to gestures, and offer assistance in a way that resembles the human behavior (Adeli et al., 2020).
- Virtual Assistant: HPE is also revolutionizing the field of virtual assistants. Through monitoring hand gestures, pose estimation enables more intuitive interactions with virtual assistants, allowing users to control smart home devices or navigate through information simply by pointing and gesturing (Kovacova et al., 2022).

This survey presents a comprehensive summary of the latest developments in Deep Learning (DL)-based 2D and 3D HPE approaches within various application domains. It explores different techniques, datasets, and evaluation metrics utilized in this field, highlighting the significant advancements and recognizing unresolved issues that require attention to enhance the precision, resilience, and utility of HPE algorithms. A clear classification of the approaches, their peculiarities, examples of use, and benchmarks are proposed to guide the reader through a comprehensive journey through the most advanced outcomes in the HPE research field.

To conduct this review, reputable databases and academic search engines were used i.e. Elsevier Scopus, Springer, IEEE Xplore, arXiv and Google Scholar. By carefully defining the keywords and phrases relevant to our topic, we collected the most cited papers published in journals or high impact conference proceedings, from 2012 until 2024. Worth to be noticed here, that we included only works that are referring to DL techniques for 2D and 3D HPE, where the pose estimation is referred to the entire human body and not to parts of it, e.g. eye- or hand-pose estimation. The next inclusion criterion was the number of citations. In order to be fair and include also the newest publications, different thresholds of the number of citations were applied, depending on the date of the publication: for papers published before 2021, the threshold was set at 50 citations, while for those published afterward, the threshold was set at 15 citations. Work on this manuscript began in early 2024, with the selection of articles carried out gradually. For recent papers, where citation counts were not a reliable criterion, we evaluated them based on quality, novelty, and the impact of their experimental results. The final selection of papers for our initial submission was completed in early June 2024, with a few additional works included on October, 2024 during the manuscript revision.

This work is organized as follows: Section 1 describes a detailed usage scenario of HPE, with challenges and opportunities, while Section 2 introduces data models to efficiently perform the different steps of HPE. Section 3 provides details about 2D and 3D approaches, for both single-person and multi-person pose estimation. The section also highlights the advantages and limitations and technical details of each method. Less popular approaches, such as radar-based and WiFi-based, are discussed as well. Section 4 briefly presents the most common frameworks, libraries, and tools for HPE, while characteristics of widely used datasets are reported in Section 5. Section 6 explains the most common metrics used to evaluate HPE approaches, along with an overview of the performance of current leaderboards on the most common datasets. Sections 7 discusses some research challenges and conclusions are drawn in Section 8.

2. Background and preliminaries

2.1. Human pose estimation

The task of HPE and tracking in computer vision involves identifying, linking, and tracing specific key points on a person's body within images or videos. These semantic keypoints include locations like elbows, knees, wrists, shoulders, hips, and ankles. Through assessing the spatial connections between these points, pose estimation algorithms can reconstruct an individual's posture, movements, and gestures. The significance of this lies in its ability to interpret the intricate language of human motion by analyzing the temporal variations in these keypoints. This technique has various applications such as recognizing simple gestures or complex activities like dancing or sports.

The human body is a complex and flexible structure that has a variety of different characteristics, such as kinematics, body shape, surface texture, etc. The choice of the model that will simulate the human body plays a crucial role in HPE and are basically three types of human body models: *skeleton-based* (Presti and Cascia, 2016), *surface-based* (Fredieu et al., 2015), and *volume-based models* (Shao et al., 2022).

Skeleton-based. It is also called as kinematic or stick figure model and despite its simplicity, it is the more commonly used human body model for both 2D and 3D HPE. In the skeleton-based model, the human body is composed of a set of articulation locations, ***keypoints***, (that are defined with their coordinates in the image) and the limb orientation (that are created by connecting the adjacent joints). Despite its obvious advantages of simple and flexible representation, it has some drawbacks, such as that it is unable to capture texture information and to represent non-rigid parts of the body (e.g., the torso) (Presti and Cascia, 2016). These keypoints, typically joints like elbows, wrists, and knees, act as a skeletal map, revealing a person's pose within an image or video. The field of HPE has some widely followed conventions and common schemas for defining keypoints. MPI (Multi-Part Inference) 16 keypoints (Andriluka et al., 2014), is a basic set that focuses on core body joints, ideal for tasks requiring general pose understanding. It includes head, neck, shoulders (left and right), elbows (left and right), wrists (left and right), hips (left and right), knees (left and right), ankles (left and right). COCO (Common Objects in Context) Keypoints (Lin et al., 2014) offer more detail, including 17 keypoints from the MPI set along with thorax (center of torso). OpenPose (Cao et al., 2017) utilizes a comprehensive set of 25 keypoints, incorporating the points from MPI and COCO along with eyes (left and right), ears (left and right), nose, center of the hip.

Surface-based. It is also called as the contour-model, and it was mostly used in earlier HPE methods, and unlike skeletal models, planar-based models emphasize the broader geometry of the body by defining it in terms of flat surfaces. Each plane corresponds to a segment of the body, such as the torso or limbs, and collectively they form a simplified, yet comprehensive, representation. While sacrificing some anatomical detail, the planar-based human body model offers a more holistic approach to capturing the overall structure and movements of the human form in various HPE applications. Widely used contour-based models include cardboard models and Active Shape Models (Fredieu et al., 2015).

Volume-based. Typically, representations of 3D human body shapes and poses involve volume-based models incorporating geometric shapes or meshes. In earlier models, geometric shapes like cylinders and conics were used to model body parts. Contemporary volume-based models are typically presented in mesh format, often obtained through 3D scans. Prominent volume-based models include Shape Completion and Animation of People (SCAPE), Skinned Multi-Person Linear model (SMPL), and a unified deformation model. Meshes are more suitable for depicting the flexible nature of the human body since they can undergo deformation (Shao et al., 2022).

3. Approaches and techniques

In this section, we delve into various techniques for estimating human poses, both in two-dimensional (2D) and three-dimensional (3D) spaces. Classical approaches, as well as recent advancements driven by deep learning, are explored. These methods address challenges such as occlusions, joint estimation accuracy, and reconstruction error. By dissecting the approaches and techniques, this section provides insights into the state-of-the-art solutions for understanding human movement and HPE strategies from visual data.

3.1. 2D human pose estimation

2D HPE aims to locate human joints from monocular videos or images. In this category, the algorithms identify and detect the critical joints of the same person in the frame, and associate them to generate the human body model. The recent Deep Learning (DL)-Based 2D HPE methods have managed to achieve greater accuracy, making the conventional technologies outdated (Salisu et al., 2023). In general, 2D HPE is divided into: single-person and multi-person Pose Estimation described in the next paragraphs. A summary of the approaches we are going to inspect is reported in Table 1.

3.1.1. 2D single-person PE

Within the realm of computer vision and HPE, the methodologies utilized are designed to accurately identify the spatial coordinates of the body joints of a specific individual captured within a frame. In instances where multiple individuals are present within the frame, a preprocessing stage becomes essential to isolate and extract the image of a *singular person*, thereby streamlining the subsequent analysis process. The complexity of the HPE task is further compounded by the varied formulations and approaches that can be adopted. In this context, the proposed methodologies that harness the power of Convolutional Neural Networks (CNNs) for facilitating the HPE task can be broadly classified into two overarching categories: *regression-based methods* and *detection-based techniques*. The regression-based methodologies focus on predicting the precise joint coordinates directly, while the detection-based strategies involve the localization of key body joints through a detection process. By categorizing these methodologies into distinct classes based on their underlying approach, it can be possible to gain a deeper understanding of the diverse techniques employed in the domain of human pose estimation using CNNs.

Regression-based methods. This large family of methods directly relates the input image to coordinates of body joints or parameters of human body models, employing a deep learning framework, based primarily on CNNs. One of the first deep neural network that was used for 2D Single-Person PE is the *AlexNet* (Krizhevsky et al., 2012), due to its simple architecture and remarkable performance. AlexNet comprises eight layers, including five convolutional layers followed by three fully connected layers. The utilization of rectified linear units (ReLU) as activation functions and overlapping pooling layers contributes to the network's efficiency in learning complex features from images. AlexNet's innovative design, utilization of data augmentation techniques, and dropout regularization played a pivotal role in revolutionizing the field of deep learning and paving the way for subsequent advancements in image classification tasks.

More recently, *DeepPose* (Toshev and Szegedy, 2014) proposes a cascade architecture of multi-stage refining regressors that are employed to refine the cropped frames from the previous stage and show improved performance. By leveraging convolutional neural networks (CNNs) and recurrent neural networks (RNNs), DeepPose can effectively capture spatial dependencies and intricate relationships between body joints for precise localization. The model's architecture allows for end-to-end training, enabling seamless integration of both feature learning and pose estimation processes.

Table 1
Big Picture of 2D HPE Approaches. For each subcategory, the symbol ♣ indicates the approach that yields the best results on the COCO dataset, ♠ for the MPII dataset, ♦ for the LSP dataset, and ♥ for the PoseTrack17 dataset.

		Paper	Model	Approach
Single-Person	Regression-Based	Krizhevsky et al. (2012)	AlexNet	CCNs + FFNs
		Toshev and Szegedy (2014)	DeepPose	CNNs + RNNs
		Carreira et al. (2016)	–	GoogLeNet
		Luvizon et al. (2019) ♠	–	Stem + BlockA + BlockB
	Detection-Based	Wei et al. (2016)	CPM	Sequential CNNs
		Newell et al. (2016)	Stacked Hourglass	CNNs + MaxPooling
		Bulat and Tzimiropoulos (2016)	–	Cascaded CNNs
		Yang et al. (2017)	–	Pyramid Residual Module
		Nibali et al. (2018)	–	CNNs + DSNT
		Tang et al. (2018), Tang and Wu (2019) ♠♦	PBN	CNNs+Hourglass+Residual Blocks
		Luo et al. (2021) ♣	–	HrHRNet + SWAHR
Multi-Person	Top-Down	He et al. (2017)	Mask R-CNN	extension of Faster-R-CNN: CNNs+Anchor Boxes+RoIAlign Layer
		Papandreou et al. (2017)	–	Faster-R-CNN + ResNet + NMS
		Chen et al. (2018)	CPN	Pyramid modules + Conv layers
		Xiao et al. (2018)	SimpleBaseline	ResNet + Deconv. Layers + NMS
		Sun et al. (2019a)	HRNet	High-to-Low Parallel Subnets
		Liu et al. (2021)	DCPose	Transformer + PTM + PRF + PCN
		Liu et al. (2022) ♥	FAMI-Pose	Transformer + Feature Extraction + Temporal Feature Alignment + Heatmap Generation
		Yang et al. (2021a) ♠	TransPose	Transformer
		Yuan et al. (2021a)	HRFormer	HRNet+Local-Window Self-Attention
		Artacho and Savakis (2021)	OmniPose	WASPv2 + Contextual Info + Gaussian Heatmap Modulation
		Xu et al. (2022) ♣	ViTPose	Vision Transformer
	Bottom-Up	Pishchulin et al. (2016)	Deepcut	Fast-R-CNN + Integer Linear Programming
		Insafutdinov et al. (2016) ♠ ♦	DeeperCut	DeepCut + ResNet + incremental optimization strategy
		Cao et al. (2017)	OpenPose	CPM + PAF
		Kreiss et al. (2019)	PifPaf	ResNet with two heads: PIF and PAF + LaplaceLoss
		Newell et al. (2017)	–	Associative Embeddings for Stacked Hourglass
		Kocabas et al. (2018) ♣	MultiPoseNet	ResNet + FPN
		Papandreou et al. (2018)	PersonLab	CNNs for each Task + Part-Induced Geometric Embedding Descriptors

Another fundamental CNN, the *GoogLeNet* (Szegedy et al., 2015) also known as *Inception* architecture, is used by Carreira et al. (2016), in the Iterative Error Feedback (IEF) network that they proposed. In this network, the estimation error is fed back into the input in a way that the network is correcting itself. GoogLeNet employs Inception modules, which allow the network to choose between multiple convolutional filter sizes within each block. These modules stack on top of each other, creating a network with 22 layers. The occasional max-pooling layers with a stride of 2 helps to reduce the grid resolution. The basic goal is to learn several levels of feature extractors throughout their joint space to derive the 2D regions of a variety of key points from an RGB image.

Luvizon et al. (2019) introduce a Soft-argmax function to convert feature maps directly to joint coordinates, enabling the learning of heat map representations indirectly without artificial ground truth generation. The proposed method utilizes a CNN with a stem, Block-A, and Block-B to provide basic feature extraction, refined features, and body-part and contextual activation maps. The approach aims to bridge the gap between detection and regression-based methods by seamlessly incorporating contextual information into the pose predictions. The study evaluates the method on challenging datasets, achieving state-of-the-art accuracy and comparable results to the state-of-the-art detection-based approaches. The proposed method is based on the Inception-v4 and Stacked Hourglass networks, and it addresses the limitations of detection-based methods by providing a fully differentiable framework and reducing computational requirements.

Detection-based methods. These methods consider body parts as targets to be detected by using two commonly employed representations: *image patches* and *heat maps of joint locations*. In conventional *Part-Based HPE methods*, body parts are initially identified from candidates of image patches. Subsequently, these detected body parts are assembled to conform to a human body model. In early work, the identified body parts tend to be relatively large and are typically depicted by rectangular sliding windows or patches. Body part detection methods often exhibit sensitivity to complex backgrounds and body occlusions. Consequently, independent image patches that rely solely on local appearance may lack the necessary discriminative features for effective body part detection (Liu et al., 2015). These methods are commonly referred to as *"top-down"*. Detected keypoints serve as the initial estimates for the pose, after obtaining these initial estimates, the method then connects the keypoints to form the complete human pose. Top-down methods typically involve pre-trained object detectors or keypoint detectors to identify body parts.

Heatmap-Based frameworks are currently prevalent in 2D "top-down" HPE methods. These frameworks typically initiate the process by regressing heatmaps. In this approach, the likelihood of a key point's presence in each pixel of an image is assessed, and a heatmap is generated to illustrate the more probable key point regions. Numerous researchers have adopted heatmap regression models, such as Convolutional Pose Machines (CPMs) proposed by Wei et al. (2016). CPMs consistently generate 2D confidence maps, employing features from the image and conviction maps from the preceding stage as inputs at each phase. The multi-stage architecture is fully differentiable, allowing training through backpropagation. Evaluation on the FLIC Dataset (Sapp and Taskar, 2013) indicates impressive performance, achieving 97.59% accuracy on elbows and 95.03% on wrists at PCK@0.2.

Newell et al. (2016) introduced a stacked hourglass network with a stacked block architecture and multiple intermediary supervisions. The name "hourglass" comes from the repeated steps of pooling and upsampling within each module. The module captures features at multiple scales by processing the input image at different resolutions. This design, incorporating convolutions in multi-level features, facilitates the re-evaluation of previous estimations and consolidate information from different spatial contexts.

Bulat and Tzimiropoulos (2016) proposed a method for human pose detection even in scenarios with significant partial occlusion. The authors introduce a CNN cascaded architecture specifically designed for learning part relationships and spatial context, and robustly inferring human pose even in cases of severe part occlusions. Their proposed detection-followed-by-regression CNN cascade effectively guides the network's focus, encodes part constraints, and leverages contextual information to predict the location of occluded body parts, achieving top performance on benchmark datasets such as MPII and LSP1 (Andriluka et al., 2014).

Yang et al. (2017) introduce a novel approach called "Learning Feature Pyramids" specifically tailored for human pose estimation, addressing the challenge of scale variations in body parts caused by changes in camera view or severe foreshortening. They noted inadequacies in existing weight initialization methods for multi-branch networks, then by learning convolutional filters on various scales of input features, obtained through different subsampling ratios in a multi-branch network, they provide a theoretical extension of weight initialization schemes to multi-branch network structures.

Unlike existing methods that either rely on heatmap matching or regress to coordinates using fully connected output layers, Nibali et al. (2018) propose a differentiable spatial-to-numerical transform (DSNT) layer. DSNT is fully differentiable, exhibits good spatial generalization, and offers a better trade-off between inference speed and prediction accuracy compared to other techniques. When replacing the popular heatmap matching approach, DSNT yields improved prediction accuracy across various model architectures.

Tang et al. (2018), Tang and Wu (2019) introduced the Deeply Learned Compositional Model for HPE, employing deep neural networks, specifically the Part-based Branching Network (PBN), to learn human body composition. Tang et al. propose a novel strategy by identifying related parts and developing specific features tailored to them. This framework features a hierarchical compositional architecture and bottom-up/top-down inference stages. Additionally, a novel bone-based part representation is proposed, efficiently encoding part orientations, scales, and shapes while mitigating large state spaces. They then introduced a multi-stage version of this network to iteratively improve intermediate features and pose estimations. Ablation experiments showed that acquiring specific features improves the location of the occluded parts, which is beneficial to HPE.

Heatmap regression commonly construct ground-truth heatmaps by covering all skeletal keypoints with 2D Gaussian kernels. Unfortunately, the fixed standard deviations of these kernels may not be suitable for bottom-up methods, which need to handle a wide range of human scales and labeling ambiguities. To address these challenges, Luo et al. (2021) recently presented the Scale-Adaptive Heatmap Regression (SAHR) method, which is capable of adjusting the standard deviation for each key point adaptively, making it more tolerant of various human scales and labeling ambiguities. This is followed by Weight-Adaptive Heatmap Regression (WAHR), contributing to a balance between foreground and background samples. Extensive testing demonstrates that combining SAHR and WAHR significantly improves the accuracy of the bottom-up assessment of human posture.

The key differences between the regression-based and detection-based methods are:

- Output Representation: The **regression-based** methods directly outputs (x, y) coordinates for each joint, while the **detection-based** methods outputs a heatmap for each joint, indicating the probability of the joint being at each pixel.
- Accuracy vs. Efficiency: The **regression-based** methods are computationally efficient but may lack precision in complex scenarios, while the **detection-based** methods are generally more accurate but require more computational resources.
- Occlusions and Complex Poses: The **regression-based** methods struggles with occlusions and complex poses because the direct regression lacks spatial context, while the **detection-based** methods are better equipped to handle occlusions and overlaps by leveraging pixel-level spatial features.
- Scalability: The **regression-based** methods do not easily generalize to multi-person scenarios, as each person's joints would require separate attention, while the **detection-based** methods can be extended to multi-person settings by associating heatmaps to different instances using additional grouping algorithms, like Part Affinity Fields or Spatial Configuration Models.

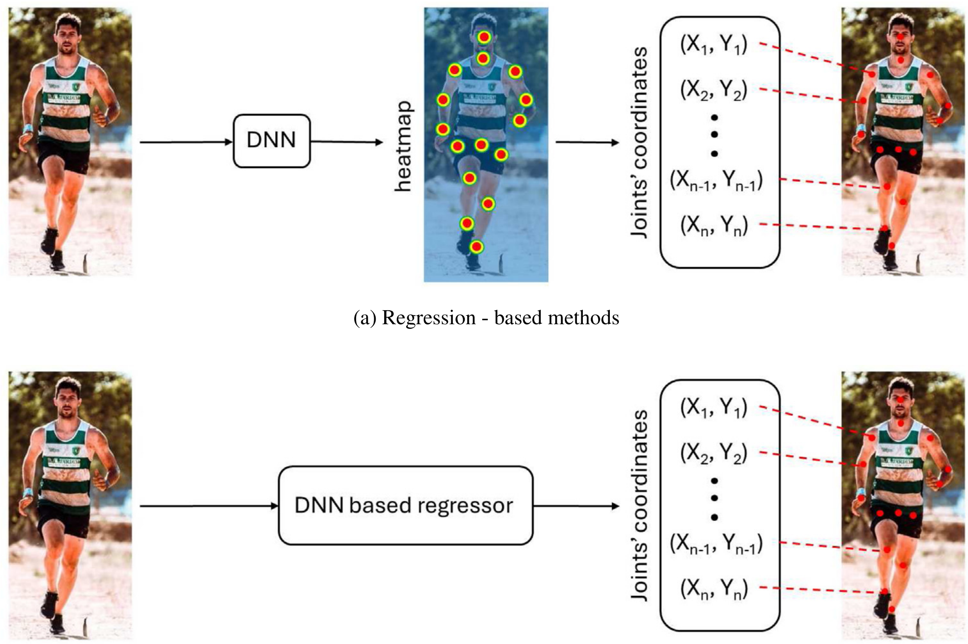
Overall, detection-based methods have become the dominant approach in HPE research due to their superior accuracy and robustness, despite their higher computational cost, while regression-based methods are still favored in scenarios where speed and computational efficiency are critical. In Fig. 1, the differences between the two methodologies are graphically shown.

3.1.2. 2D multi-person PE

Differing from the task of estimating poses for a single-person, multi-person PE faces the additional challenge of handling both detection and localization in space tasks. This is because there is no indication of the number of individuals present in the input images and where they are spatially placed. In a multi-person scenario, top-down methods typically utilize person detectors to identify the bounding boxes of each individual in the input image. Subsequently, these methods employ existing single-person PE methods to predict the poses of each person. The accuracy of person detection significantly influences the precision of the predicted poses, and the overall system's runtime is directly proportional to the number of persons detected. Conversely, bottom-up methods directly predict the 2D joint locations for all individuals in the image and then assemble these joints into independent skeletons. Achieving the correct grouping of joint points in complex environments poses an intriguing research problem for bottom-up methods.

Top-down approaches. The top-down approach in HPE operates through different stages. Initially, it identifies the person within a visible frame and then extracts the recognized object. Finally, the technique estimates a specific number of crucial points for each identified person within that frame before constructing the skeletal structure.

He et al. (2017) introduced a conceptually flexible, and general framework called Mask R-CNN for object instance segmentation, efficiently detecting objects in an image while simultaneously generating high-quality segmentation masks for each instance. The method is based on the extension of the Faster-R-CNN (Ren et al., 2017) model. Anchor boxes (also known as default boxes) are predefined bounding



(a) Regression - based methods

(b) Detection - based methods

Fig. 1. Comparison of (a) regression-based and (b) detection-based methods.

boxes of various sizes and aspect ratios. They serve as reference templates for detecting objects at different scales and shapes within an image. Objects can vary significantly in size and proportions. Anchor boxes allow the model to adapt to this variability. Faster R-CNN employs a set of anchor boxes at each spatial position in the feature map. Each anchor box has different scales (sizes) and aspect ratios (width-to-height ratios). The network predicts whether an anchor contains an object (foreground) or not (background). In Faster R-CNN, there are typically 9 anchors at each position. They play a critical role in achieving accurate and robust object detection. He et al. connect keypoint heads onto Region of Interest Aligned (RoIAlign Layer) feature maps, to create an individual one-hot mask for every keypoint that preserves exact spatial locations. The method achieves optimal results across multiple tasks including instance segmentation, bounding-box object detection, and person keypoint detection.

Papandreou et al. (2017) proposed a two-stage top-down approach for person detection and keypoint estimation. In the first stage, they used Faster-R-CNN to predict likely bounding boxes containing people. In the second stage, they employed a fully convolutional ResNet to estimate keypoints within each proposed box, generating dense heatmaps and offsets. Their novel aggregation procedure combined these outputs for highly localized keypoint predictions, while keypoint-based Non-Maximum-Suppression and confidence score estimation improved accuracy compared to traditional box-level methods. Non-Maximum Suppression (NMS) is a crucial technique used to filter out redundant or overlapping predictions. It involves sorting bounding boxes by confidence, selecting the highest-confidence box, and then iteratively removing overlapping boxes based on a predefined threshold, resulting in cleaner and more accurate results.

Chen et al. (2018) reused the concept of "Learning Feature Pyramids" proposed by Yang et al. (2017) by designing a Cascade Pyramid Network (CPN) that utilizes multi-scale feature maps from various layers to extract increased information from both local and global features. CPN consists of a series of pyramid modules, each capturing features at different scales, then these pyramid modules are cascaded to refine pose estimation progressively. CPN constructs feature pyramids by applying convolutional layers with different receptive fields. These pyramids capture multi-scale features, allowing the network to handle keypoints at varying resolutions. Each pyramid module predicts heatmaps for individual keypoints. These heatmaps represent the likelihood of each keypoint's presence at different image locations. CPN introduces intermediate supervision by attaching supervision (loss) at each pyramid module. This encourages the network to learn better features at different scales during training.

Xiao et al. (2018) introduced a straightforward pose estimation network simply adding a few deconvolutional layers over the last convolution stage in the ResNet model (He et al., 2016), called C5. For the processing frame in videos, the boxes from a human detector and boxes generated by propagating joints from previous frames using optical flow are unified using a bounding box Non-Maximum Suppression (NMS) operation. The boxes generated by propagating joints serve as the complement of missing detections of the detector.

Sun et al. (2019a) started from a high-resolution subnetwork as the first stage, gradually add high-to-low resolution subnetworks one by one to form more stages and connect the multiresolution subnetworks in parallel. The proposed architecture, High-Resolution Net (HRNet), stands out for its ability to maintain high-resolution representations throughout the entire process of human pose estimation. By connecting high-to-low subnetworks in parallel and conducting repeated multi-scale fusions, HRNet achieves spatially precise heatmap estimation without the need for a separate low-to-high upsampling process. This approach leads to superior keypoint detection accuracy, efficiency in computation complexity, and parameter efficiency compared to existing methods. Additionally, HRNet outperforms traditional networks by generating reliable high-resolution representations and conducting information exchange across parallel multi-resolution subnetworks.

When in 2017 Transformers architectures took hold in the world of scientific research in AI (Vaswani et al., 2017), numerous approaches for HPE adopted this model to improve their performance. The basic knowledge, later used in these approaches is provided into the work of Liu et al. (2021, 2022)

Liu et al. (2021) introduced DCPose, a novel multi-frame HPE framework that utilizes temporal cues for keypoint detection. The framework leverages abundant temporal cues between video frames to facilitate keypoint detection, addressing challenges in handling motion blur, video defocus, and pose occlusions. The framework comprises three components: (i) a Pose Temporal Merger (PTM) for spatiotemporal context encoding, implemented via a group convolutional layer; (ii) a Pose Residual Fusion (PRF) module for computing weighted pose residuals using basic dense layers; (iii) a Pose Correction Network (PCN) for refining pose estimations composed by Deformable Convolution V2 Networks (DCN v2) (Zhu et al., 2019). Liu et al. (2022) investigated also the task of multi-frame HPE by focusing on the efficient utilization of temporal contexts through feature alignment and complementary information extraction. They introduced a hierarchical coarse-to-fine network designed to align supporting frame features progressively with key frame features, while a mutual information objective to provide effective supervision on intermediate features is presented. The proposed method achieves state-of-the-art results on three benchmark datasets: *PoseTrack2017* and *PoseTrack2018* (Andriluka et al., 2018), *Sub-JHMDB* (Jhuang et al., 2013).

By following the growing interest in transformer-based approaches, due to the capabilities of attention layers to capture long-range dependencies and global information of the predicted keypoints, Yang et al. (2021a) introduced *TransPose*. The attention layers in Transformer efficiently capture long-range relationships, revealing dependencies of predicted keypoints. The last attention layer serves as an aggregator for predicting keypoint heatmaps, following the principle of Activation Maximization, offering fine-grained, image-specific dependencies that maximize the activation of specific neurons in the network. TransPose achieves 75.8 and 75.0 Average Precision (AP) on *COCO* (Lin et al., 2014) validation and test-dev sets, respectively. It results as lightweight and faster than mainstream CNNs, with superior performance on MPII benchmark (Andriluka et al., 2014) when fine-tuned at low training costs.

Yuan et al. (2021a) introduced *HRFormer*, a High-Resolution Transformer designed for dense prediction tasks where the model should produce predictions for every pixel in the input image. By learning high-resolution representations and leveraging a multi-resolution parallel design from HRNet, along with local-window self-attention, HRFormer addresses the high memory and computational costs associated with low-resolution representations. This self-attention mechanism performs attention over small, non-overlapping image windows improving memory and computation efficiency. The model, by including a convolution step in the feed-forward layer for information exchange, demonstrated effectiveness in HPE and semantic segmentation tasks on the COCO dataset (Lin et al., 2014). Similarly to high-resolution convolutional networks (HRNet) (Sun et al., 2019a), HRFormer can handle multi-resolution features effectively.

OmniPose (Artacho and Savakis, 2021) architecture utilized a novel “Waterfall Atrous Spatial Pyramid” module, or WASPv2, to enhance multi-scale feature representations, improving backbone feature extractor efficacy without postprocessing. WASPv2 module expands the feature extraction through its multi-level architecture. It increases the FOV (Field-of-View) of the network with consistent high resolution processing of the feature maps in all its branches, which contributes to higher accuracy. FOV refers to the angular extent of the observable world that is seen at any given moment. In addition, WASPv2 generates the final heatmaps for joint localization without the requirement of an additional decoder module, interpolation or pooling operations. By incorporating contextual information and Gaussian Heatmap Modulation introduced by the authors, OmniPose achieved state-of-the-art accuracy in HPE, maintaining efficient multi-scale representations with progressive filtering and comparable field-of-view to spatial pyramid configurations.

Xu et al. (2022) introduced *ViTPose*. It leverages plain and non-hierarchical Transformer architecture as backbone, notable for its simplicity, scalable size (100M to 1B parameters), and flexibility in training. ViTPose achieves a new Pareto front between throughput and performance, demonstrating versatility in attention type, input resolution, pre-training, finetuning strategy, and addressing multiple pose tasks, with empirical results showing successful knowledge transfer from large to small models, and the largest model setting a new state-of-the-art (81.1 AP on the COCO test-dev set (Lin et al., 2014)). Code and models are available on their GitHub Webpage: https://github.com/ViTAE-Transformer/ViTPose

Bottom-up approaches. Bottom-up approaches employ a shared global detector to initially identify all potential joints, which are then grouped into distinct human instances based on a specific artificially defined joint affinity method. Many algorithms address these two stages independently.

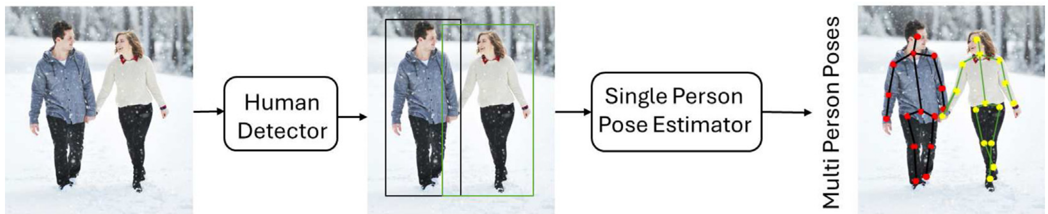
In the case of *DeepCut* (Pishchulin et al., 2016), a Fast-R-CNN-based (Girshick, 2015) body part detector is utilized to initially identify all potential body part candidates. Subsequently, each part is assigned a label corresponding to its category, and these labeled parts are then assembled into a complete skeleton using integer linear programming. Building upon DeepCut, *DeeperCut* (Insafutdinov et al., 2016) enhances the methodology by employing a more robust part detector based on *ResNet model* (He et al., 2016) and an improved incremental optimization strategy that explores geometric and appearance constraints among joint candidates. DeeperCut introduces novel pairwise terms that consider image context. These terms allow assembling body part proposals into consistent configurations. By conditioning on the image, DeeperCut achieves more accurate grouping of body parts. Moreover, DeeperCut optimizes the pose estimation process more efficiently. It explores the search space incrementally, solving for reliable body parts first. This strategy leads to better performance and significant speed-up factors.

OpenPose (Cao et al., 2017) takes a different approach by using CPM (Convolutional Pose Machines) (Wei et al., 2016) to predict candidates for all body joints through Part Affinity Fields (PAFs). PAFs predict the vector fields connecting body parts. These vectors guide the association of joints to form complete limbs. For instance, the PAF between shoulder and elbow guides the arm’s orientation. The proposed PAFs effectively encode the locations and orientations of limbs, facilitating the assembly of estimated joints into various person poses. It represents the first real-time multi-person system capable of jointly detecting 135 keypoints for human body, hand, facial, and foot estimation in single images, and it achieves this by maintaining runtime invariance to the number of detected people.

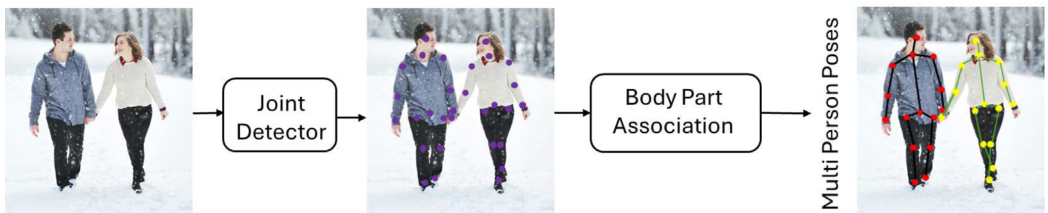
Similarly, following a strategy akin to OpenPose, Kreiss et al. (2019) introduce a PifPaf net designed to predict a Part Intensity Field (PIF) and a PAF, representing body joint locations and associations. PIFs localize individual body parts (such as joints) in the image. They generate heatmaps that represent the likelihood of a specific body part being present at a particular location. For example, a PIF heatmap for the wrist indicates the wrist’s estimated position. PAFs encode the pairwise associations between body parts. Notably, it performs well on low-resolution images due to the finely detailed PAF and the incorporation of Laplace loss. It achieves real-time performance while maintaining high accuracy and results particularly well-suited for urban mobility scenarios like self-driving cars and delivery robots. Its performance makes it effective in challenging environments with low-resolution images and multiple people.

The mentioned approaches all adhere to a separation between joint detection and joint grouping. However, more recently, there have been methods that can predict both in a single stage. Newell et al. (2017) introduced a unified DNN architecture that can concurrently handle both detection and grouping. The proposed method supervises CNNs for both detection and grouping and simultaneously outputs detections and group assignments using a joint predictive loss. This approach seamlessly integrates into state-of-the-art network architectures. The network can produce heatmaps for detecting each joint, as well as embedding maps that contain grouping tags for each joint.

Kocabas et al. (2018), integrates a multi-task model with a novel assignment method to address human keypoint estimation, detection, and semantic segmentation collectively, named MultiPoseNet. Its backbone network is a fusion of ResNet model and FPN (Feature Pyramid



(a) Top - down methods



(b) Bottom - up methods

Fig. 2. Comparison of (a) top-down and (b) bottom-up methods.

Network) with shared features for keypoints and person detection subnets. The results of human detection serve as constraints for the spatial positioning of individuals.

Papandreou et al. (2018) introduced PersonLab, a box-free bottom-up approach for PE and instance segmentation in multi-person images. It avoids bounding boxes and directly predicts keypoints and instance masks operating in a single-shot manner for accurate results. The model combines part-based modeling with a convolutional network to detect keypoints and predict their displacements, enabling grouping into person pose instances. PIFs predict the likelihood of keypoints (e.g., joints) being present at specific locations. Heatmaps represent the confidence of each keypoint's position. PAFs encode pairwise associations between keypoints. They predict vector fields connecting keypoints to form complete poses. Additionally, they proposed a part-induced geometric embedding descriptor for associating semantic person pixels with their respective instances, achieving efficient instance-level segmentation regardless.

The key differences between the top-down and bottom-up methods are:

- Processing Order: The **top-down** methods detect first the persons, followed by keypoint estimation for each detected person, while the **bottom-up** methods detect all keypoints first, followed by person-level association (Fig. 2).
- Accuracy vs. Efficiency: The **top-down** methods generally achieve higher accuracy for each individual's pose due to the focus on cropped regions, but are less efficient and harder to scale, while the **bottom-up** methods are more efficient for multi-person scenarios but may sacrifice some accuracy, especially in dense crowds.
- Scalability: In the **top-down** methods, the complexity increases linearly with the number of people because each person requires a separate HPE pass, while in the **bottom-up** methods, the complexity remains relatively constant regardless of the number of people, since joint detection is performed in a single pass over the whole image.
- Sensitivity to Initial Detection: The **top-down** methods are highly dependent on the quality of the initial person detection step, since missing detections directly lead to missing poses, while the **bottom-up** methods are robust to missing persons but may struggle with keypoint association, leading to potential mixing of joints between people.
- Post-Processing Requirements: In the **top-down** methods, minimal post-processing is needed since each person is handled independently, while the **bottom-up** methods require complex post-processing to group detected keypoints accurately into individual human skeletons.

Overall, top-down methods are favored in applications where pose accuracy is crucial, such as sports analysis or detailed gesture recognition, where individuals are processed separately and high precision is required. However, they may not be practical in large crowds or when real-time performance is needed. Bottom-up methods are typically chosen for real-time multi-person scenarios such as crowd analysis, surveillance, or large-scale human behavior monitoring, where efficiency and the ability to process multiple people in one pass are critical. The choice between top-down and bottom-up approaches depends heavily on the specific requirements of the application, such as the number of people to be detected, the desired level of accuracy, and the available computational resources.

3.2. 3D human pose estimation

Recently, there has been a growing interest in the research field of 3D HPE, owing to its extensive applicability. However, this area is far from being completely addressed due to its distinctive challenges compared to 2D HPE. In 2D, challenges primarily involve variations in body poses, complex backgrounds, diverse clothing appearances, and occlusions. On the other hand, 3D HPE encounters additional obstacles, including the absence of comprehensive 3D datasets in real-world scenarios, challenges related to depth ambiguities, a substantial need for detailed posture information, including rotations and translations, and the exploration of a vast state space for each joint, which represents a discretized 3D space. In the following section, we will discuss the most common strategies to tackle these challenges and obtain satisfactory results. A recap of the investigated approaches and results over the most common datasets of 3D HPE are reported in Section 6.3, Tables 7, 8, 9. A summary of the approaches we are going to inspect is reported in Table 2. Note that for Multi-Person approaches, no direct comparison is provided due to the lack of comparable results for these models. In some cases, such as with SMPL models, it may appear that an older model outperforms the one developed five years later, however, this impression arises because performance data for the newest models are not included. This exclusion is due either to uncertainty regarding the consistency of evaluation protocols across models or to discrepancies in reported results across the papers citing these models.

Table 2
Big Picture of 3D HPE Approaches. For the subcategories compared in the tables in Section 6.3, the symbol ♣ indicates the approach that yields the best results on the Human3.6M dataset with the P1 protocol, ♠ for the Human3.6M dataset with the P2 protocol, and ♦ for the MPI-INF-3DHP dataset.

		Paper	Model	Approach
Single Person	Direct Prediction of 3D Poses	Pavlakos et al. (2017a)	–	CNNs + Volumetric Representation + with intermediate supervision and iterative estimation
		Sun et al. (2017)	–	Reparameterized pose representation using bones instead of joints + compositional loss function
		Sun et al. (2018)	Integral	CNNs + FFN + Integral Regression
		Luvizon et al. (2021) ♣	–	CNNs + pose regression + multi-scale pose + action supervision + re-injection
		Choi et al. (2021) ♠	MobileHumanPose	MobileNetV2 + parametric activation function + skip concatenation from U-Net
		Wang et al. (2024)	Di^{2}Pose	Compositional pose representation + Diffusion model
	Lifting from 2D Poses	Güler et al. (2018)	DensePose	Dense Regression + Mask-RCNN
		Habibie et al. (2019)	–	CNN predicting root centered 3D pose + CNN predicting viewpoint parameters
		Zhou et al. (2019) ♣ ♠	HEMlets	Three joint-heatmaps + CNN + volumetric joint-heatmap regression
		Zhao et al. (2019)	SemGCN	Regression + Graph-structured data
		Ci et al. (2019)	LCN	GCN and FCN
		Chen et al. (2019b)	–	Unsupervised-learning: 2Dto3D + 3Dto2D
		Wang et al. (2019b)	NRSfM	Loss function that ties depth prediction
		Li et al. (2020a)	TAG-Net	Data augmentation
		Zeng et al. (2020)	SRNet	Body local regions + separate network branches
		Gong et al. (2021)	PoseAug	Pose augmentor + feedback from estimation + Kinematic Chain Space
		Xu and Takano (2021)	GraphSH	Graph Stacked Hourglass Networks
		Zou and Tang (2021)	OriNet	Weight modulation + Affinity modulation
		Jiang et al. (2024)	ZeDO	Optimization pipeline + Diffusion Model
		Du et al. (2024) ♦	JoyPose	data augmentation with reinforcement learning + anatomy-aware global-local pose feature representation
	SMPL Models	Kanazawa et al. (2018) ♠	HMR	Minimize reprojection + adversarial component
		Pavlakos et al. (2018b)	PosePrior ShapePrior	CNNs + PosePrior + ShapePrior + 3D per-vertex loss + differentiable renderer
		Omran et al. (2018)	NBF	CNNs + image partitioning + probability maps + SMPL in 2D + loss on vertex positions
		Tripathi et al. (2023)	IPMAN	Pressure heatmap + CoP + CoM
	Multi-View Images	Pavlakos et al. (2017b)	–	2D joint heatmaps + 3DPS + CNNs
		Qiu et al. (2019)	–	CNNs + cross-view fusion scheme + 3D Pictorial Structure KTPFormer
		Kocabas et al. (2019)	EpipolarPose	Epipolar Geometry
		Zhang et al. (2021)	AdaFuse	Adaptive fusion weights for each camera + heatmap sparsity
		Wandt et al. (2021)	CanonPose	Multi-view consistency constraints
		Reddy et al. (2021)	TesseTrack	4D CNN for short-term individual repr. + differentiable matcher into 3D poses
		Chun et al. (2023b) ♣	–	VHA + BCP
	Sequence of Monocular Images	Hossain and Little (2018)	–	Layer-normalized LSTM units + shortcut connections + temporal smoothness
		Dabral et al. (2018)	Temporal PoseNet	Two-layer FCN + temporal information + 2D joint coordinates
		Xu et al. (2018)	MonoPerfCap	CNNs + batch-based pose estimation
		Cheng et al. (2019)	–	Occlusion-aware 2D temporal CNN + Cylinder Man Model
		Cai et al. (2019)	–	2D pose sequence in the form of a graph + Local-to-Global network

(*continued on next page*)

Table 2 (*continued*).

Single Person	Sequence of Monocular Images	Li et al. (2019a)	–	Additional train on previous annotations to learn new pose
		Hu and Ahuja (2021)	HDVR	LSTM + 3D movement subsequences
		Li et al. (2021)	Lifting Transformer	Strided Transformer Encoder
		Yuan et al. (2021b)	SimPoE	Reinforcement learning + kinematic pose refinement
		Zheng et al. (2021)	PoseFormer	Spatial–temporal Transformer
		Hu et al. (2021)	–	Spatial–temporal directed Graph Convolution + U-shaped network
		Li et al. (2022b)	MHFormer	Cascaded Transformer-based architecture + models multiple plausible pose hypotheses
		Zhang et al. (2022) ♠	MixSTE	Temporal Transformer block + Spatial Transformer block
		Gholami et al. (2022)	AdaptPose	Adversarial training approach for cross-dataset generalization
		Zhu et al. (2023) ♣	MotionBERT	Motion encoder from 2D to 3D motion with geometric, kinematic, and physical insights
		Zhao et al. (2023)	PoseFormerV2	Compact representation of lengthy skeleton sequences
		Mehraban et al. (2024)	MotionAGFormer	GCNFormer + Transformer
		Peng et al. (2024) ♦	KTPFormer	kinematic relationships and topology + trajectory topology
		Zhang et al. (2024)	APP module	2D feature maps + image and temporal features
	Sequence of Multi-View Images	Rhodin et al. (2016)	–	Gaussian Density field + Edge-based Alignment Energy
		Huang et al. (2017)	–	Alignment of 2D features to 3D body models + CNNs to segment the individual+ DCT temporal prior
		Trumble et al. (2018)	–	Symmetric Convolutional Autoencoder + dual loss
Multi-Person	Single Monocular Image	Mehta et al. (2018)	–	Occlusion-Robust Pose-Maps
		Moon et al. (2019)	–	3D human root localization + root-relative 3D single-person PE
		Nie et al. (2019)	SPM	Structured Pose Representation
		Rogez et al. (2020)	LCR-Net	CNNs + pose proposal generator + classifier + regressor from 2D to 3D
	Multi-View Images	Dong et al. (2019)	–	3DPS + multi-way matching algorithm to cluster detected 2D poses
		Rhodin et al. (2019)	–	Spatial layout + instance segmentation + and body representation
		Dong et al. (2021)	–	Optimization scheme for multi-view 2D observations and 3D joint candidates
	Sequence of Monoc. Imgs	Zanfir et al. (2018)	–	Feedforward predictors + semantic fitting
	Sequence of Multi-View Imgs	Belagiannis et al. (2014b)	–	3DPS + temporal coherence (multi-view human tracking)

3.2.1. 3D single-person pose estimation

The simplicity of image processing requirements fuels efforts to extract a 3D human pose from a single image, but they face a challenge where different 3D poses can correspond to the same 2D images. Attempts to use temporal or multi-view information for clarity prove impractical in the recovery process.

Extensive research has led to the development of diverse methods for tackling these challenges. In the following, these methods are introduced and discussed, focusing on six main categories for 3D single-person pose estimation based on a single monocular image: *direct prediction of 3D poses*, *lifting from 2D poses*, *SMPL model*, *Multi-View Images*, *Sequence of Monocular Images*, and approaches based on the *Sequence of Multi-View Images*.

Direct prediction of 3D poses. The simplest approach for 3D HPE involves creating an end-to-end network that predicts the 3D coordinates of joints in poses. Techniques directly translating input images into 3D body joint positions fall into two main categories: *detection-based methods* and *regression-based methods*. It is noteworthy that efforts have been made to integrate the heatmap representation and joint regression.

Detection-Based Techniques anticipate a probability heatmap for each joint, with the joint's position established by identifying the highest likelihood point on the heatmap. Pavlakos et al. (2017a) use a volumetric representation and CNN Network to predict voxel-wise joint likelihoods, employing a coarse-to-fine strategy with intermediate supervision and iterative estimation to enhance resolution in 3D pose estimation. Voxel-wise joint likelihoods essentially represent the probability of observing a specific value (intensity or signal) within a voxel (3D Cubic Data Representations), considering information from multiple sources of information for the same voxel location.

HPE, especially for 2D approaches, relies on *Regression-Based Methods*. They directly predict the positions of joints in relation to the root joint location. Sun et al. (2017) advocate for bone regression in pose estimation, citing the ease of learning, enhanced geometric constraint capture, and stability of bone representations, but note the need to convert pose data to a relative bone-based format due to the local and independent nature of L2 loss for bones. The authors proposed a structure-aware regression approach. It adopts a reparameterized pose representation using bones instead of joints. To combine the advantages of both detection and regression-based approaches, Sun et al. (2018) suggest a straightforward and efficient integral regression method. In this approach, the joint position is predicted as the probability-weighted average of all positions in the heatmap. The model consists of a deep CNNs backbone network to extract convolutional features from the input image and a shallow FFN network to estimate the target output (heat maps or joints) from the features. The authors show that the integral regression method is a flexible component that can be easily embedded into various backbone networks, and the result is less affected by the network capacity than the heat map. This technique facilitates end-to-end training while demanding minimal computation and storage.

Luvizon et al. (2021) introduced a precise 3D pose estimation method using low-resolution feature maps, avoiding the need for volumetric heat maps. Their adaptable approach, combining CNN architecture and pose regression, achieves efficient multi-scale supervision, benefiting from both 2D and 3D data and showing significant improvements in 3D pose estimation, with support for seamless training on single frames and video clips simultaneously. Choi et al. (2021) contend that lightweight backbones have poor performance in pose estimation. To address this, they introduced MobileHumanPose, a mobile-friendly model utilizing a modified MobileNetV2 backbone (Sandler et al., 2018), parametric activation function, and skip concatenation from U-Net (Ronneberger et al., 2015), achieving comparable performance to leading models with a model size seven times smaller than ResNet-50 (He et al., 2016).

Recently, a new approach that differs from *detection-based* and *regression-based* existing methods has been recently widely adopted, that is based on *diffusion-based* models. The advantage of this approach, as observed in other tasks (Ho et al., 2020, 2022; Dhariwal and Nichol, 2021; Avrahami et al., 2022; Huang et al., 2022; Brempong et al., 2022; Gu et al., 2022; Chen et al., 2023; Yang et al., 2023; Kong et al., 2023), is the ability to handle complex and uncertain data distribution. Diffusion Model is a latent variable model which maps to the latent space using a fixed Markov chain, which gradually adds noise to the data. After the fixed steps of the Markov chain, the data are transformed to pure Gaussian noise. In the training phase, a diffusion model aims to learn the reverse process, i.e. generating additional data starting from the pure Gaussian noise. Typically, in the HPE task, the use of a Diffusion Model, involves an approach which progressively refines the pose distribution in a continuous space (Gong et al., 2023a; Choi et al., 2023; Feng et al., 2023; Zhou et al., 2023), which may requires a larger search space to achieve optimal generative outcomes (Gong et al., 2023b; Xie et al., 2023; Austin et al., 2021). A discrete diffusion model for occluded 3D HPE is proposed by Wang et al. (2024). This work, named Di^{2}Pose consists on a method that introduces occlusion in the latent space without extra priors or explicit augmentations, providing a deeper feature-based understanding of occlusion's effects on pose estimation. Specifically, Di^{2}Pose employs a two-stage approach: a pose quantization step followed by a discrete diffusion process. The pose quantization step leverages the discrete nature of 3D poses and represents them as quantized tokens by capturing the local interactions between joints. Subsequently, the discrete diffusion process models the quantized pose tokens in the latent space through a conditional diffusion model. The model demonstrated to be highly performing and efficient.

Lifting from 2D poses. Numerous studies aimed to leverage the outcomes of 2D PE to enhance the generalization performance of 3D HPE in diverse real-world scenarios, motivated by the swift progress in 2D HPE algorithms. Habibie et al. (2019) bridge the gap between 2D and 3D pose estimation, demonstrating the potential for accurate and robust human pose recovery in real-world scenes. They propose a novel deep learning architecture that leverages both 2D and 3D cues. Their method disentangles hidden space encoding for explicit 2D and 3D features, allowing joint training on image data with 3D labels and image data with only 2D labels. By incorporating a learned projection model from predicted 3D pose, their approach achieves state-of-the-art accuracy on challenging in-the-wild data, i.e. MPI-INF-3DHP (Mehta et al., 2017a). Part-centric heatmap triplets (HEMlets) (Zhou et al., 2019) bridge the gap between 2D observations and 3D interpretations. HEMlets, by dividing the full-body skeleton into 14 parts, captures depth information through the use of three joint-heatmaps: negative, zero, and positive ones. Their approach involves training a CNN to predict body parts from input images, followed by volumetric joint-heatmap regression. The integral operation extracts joint locations from volumetric heatmaps, ensuring end-to-end learning. Despite its simplicity, their network outperforms state-of-the-art methods by approximately 20% on the Human3.6M dataset (Ionescu et al., 2014). Notably, their method supports training with "in-the-wild" images, where only weakly annotated relative depth information is available, enhancing generalization.

Leveraging geometric connections among human pose joints is instrumental in HPE algorithm design. Zhao et al. (2019) propose the Semantic Graph Convolution Network (SemGCN), treating the human pose skeleton as a graph to learn edge weights, while (Ci et al., 2019) introduce the locally connected network (LCN), using distinct filters for feature extraction and addressing the limitations of the natural skeleton adjacency matrix. Chen et al. (2019b) use an unsupervised approach, exploiting rotation invariance in the generated 3D pose. They convert a 2D pose to 3D, compute the loss between two 3D poses (one after a rotation step), and then transform the 3D pose back to 2D to calculate the loss against the original 2D input.

Adversarial Learning has been widely employed for generating life-like 3D poses. This kind of learning strategy involves a game-like scenario between two neural networks: the generator and the discriminator. The generator generates images that look like the original photo. The discriminator has to distinguish between real data (e.g., actual person photos) and fake data (e.g., generated person-like images). This helps to create more robust models by making them resilient to deceptive inputs and improving their overall performance. Güler et al. (2018) introduced "DensePose", a dense HPE model, by establishing correspondences between RGB images and a surface-based human body representation. Efficiently annotating 50,000 individuals from the COCO dataset (Lin et al., 2014), the researchers train CNN-based systems for *in-the-wild* scenarios, enhancing effectiveness through an inpainting network that can fill in missing ground truth values, and achieving real-time, highly accurate results with region-based models (Mask R-CNN (He et al., 2017)).

Wang et al. (2019b) propose a novel approach to 3D human pose estimation that leverages knowledge from Non-Rigid Structure from Motion (NRSfM). Unlike traditional supervised methods that require abundant 3D data or multi-view footage, their method relies solely on 2D landmark annotations, alleviating the data bottleneck. NRSfM methods reconstruct 3D shapes and camera positions from multiple 2D projections of articulated 3D points. These points do not have to belong to the same object, but can be from multiple instances of the same object category, which naturally applies to the specific problem. The challenge lies in using NRSfM as a teacher, as it often produces inaccurate depth reconstructions when faced with strong 2D projection ambiguities. Instead of directly using these erroneous depth estimates as hard targets, the authors introduce a novel loss function that ties depth prediction to the cost function used in NRSfM. This allows the pose estimator to learn from image features and reduce depth errors. Their method adopts a 3D shape dictionary to retrieve camera matrices and codes, facilitating the reconstruction of depth information. Validated on the Human3.6M dataset (Ionescu et al., 2014), their learned 3D pose estimation network achieves more accurate reconstruction compared to NRSfM methods and outperforms other weakly supervised approaches despite using significantly less supervision.

Li et al. (2020a) presented a scalable data augmentation method, TAG-Net, creating a large training dataset, i.e., U3DPW, (over 8 million

3D human poses with corresponding 2D projections) for 2D-to-3D network training, effectively addressing dataset bias. The approach transforms a limited dataset by synthesizing novel 3D human skeletons using hierarchical representation and heuristics, resulting in state-of-the-art accuracy on a major benchmark and superior generalization to unseen and rare poses, as demonstrated through extensive experiments.

Zeng et al. (2020) proposed SRNet, a method that divides the body into local regions processed in separate network branches, maintaining global coherence by reintegrating context. This split-and-recombine approach, adaptable to single-image and temporal models, significantly improves predictions, especially for rare and unseen poses, by focusing on local pose statistics without sacrificing essential information for joint inference.

Gong et al. (2021) proposed PoseAug, an innovative pose enhancement tool that modifies posture, body dimensions, viewpoint, and position using differentiable techniques. PoseAug works in tandem with a 3D pose estimator to create varied and complex poses, guided by feedback from estimation errors. It introduces a Part-aware Kinematic Chain Space (KCS) to evaluate the credibility of joint angles, enhancing pose diversity while maintaining plausibility. These features enhance the adaptability of pose estimators across different methods of 3D pose estimation.

Xu and Takano (2021) proposed the Graph Stacked Hourglass Networks, a novel Graph Stacked Hourglass Networks (GraphSH) with repeated encoder–decoder structures for 3D HPE. This architecture processes graph-structured features at three scales, enabling the learning of essential local and global representations, and introduces a multi-level feature learning approach for improved performance.

Zou and Tang (2021) address limitations in prior graph convolutional network (GCN) methods (Zhang et al., 2019), which suffer from shared feature transformations for each node within a graph convolution layer and suboptimal graph definitions based on the human skeleton. To overcome these drawbacks, they introduce a novel Modulated GCN that consists of two key components: weight modulation and affinity modulation. Weight modulation disentangles feature transformations for different nodes, enhancing model flexibility while maintaining a compact size. Affinity modulation adjusts the graph structure to model additional edges beyond the natural connections of body joints. Rigorous ablation studies demonstrate the effectiveness of both types of modulation, resulting in improved performance with minimal overhead. Compared to state-of-the-art GCNs for 3D human pose estimation, their approach significantly reduces estimation errors or drastically reduces model size while achieving comparable performance.

Jiang et al. (2024) propose a Zero-shot Diffusion-based Optimization (ZeDO) pipeline for 3D HPE to solve the problem of cross-domain and in-the-wild 3D HPE. The pipeline combines a simple yet effective optimization pipeline with a pre-trained diffusion-based 3D human pose generation model. The diffusion model denoises the output from the optimization pipeline iteratively to ensure optimized poses following human body constraints. Meanwhile, poses are optimized by minimizing 2D keypoint re-projection errors with a simple yet effective optimization pipeline, i.e. pose rotation to an optimal pose.

Du et al. (2024) propose JoyPose data augmentation and HPE as joint tasks. They assume mutual promotion between them would be yielded through joint learning, and further propose a joint training framework that consists of evolutionary data augmentation and global-local representation for 3D HPE. The evolution for data augmentation is guided by a reinforcement learning strategy in a probabilistic way according to pose estimation loss. The anatomy-aware global-local pose feature representation module separately captures global features and local features according to anatomical and kinematic patterns observed from pose estimation errors across different human joints.

SMPL models. The reconstruction of human meshes often relies on volumetric models, which provide additional information about the intricacies of the human form. A mesh refers to a collection of vertices, edges, and faces that define the shape of a 3D object. Essentially, a mesh is a simplified representation of a physical object that can be used to create digital models for use in computer simulations, and other applications. The SMPL model (Skinned Multi-Person Linear) is a widely recognized and popular volumetric model for 3D HPE. It is compatible with established rendering engines and has been embraced by many. What makes the SMPL model stand out is its ability to incorporate pre-existing knowledge about human shape, making it easy to adapt with minimal data. Moreover, various optimization techniques have been suggested for the estimation of 3D human pose, further enhancing the model's accuracy and reliability. With its versatility and proven track record, the SMPL model is undoubtedly one of the most effective approaches in the literature.

Recent approaches employ diverse neural networks to predict SMPL parameters across various tasks directly. For instance, Kanazawa et al. (2018) introduce an end-to-end deep learning scheme aiming to understand the mapping from image pixels to SMPL model parameters known as Human Mesh Recovery (HMR). Unlike most existing approaches that compute 2D or 3D joint locations, HMR produces a richer and more informative mesh representation that is parameterized by both shape and 3D joint angles. The primary objective is to minimize the reprojection loss of keypoints, allowing the model to be trained using images "in-the-wild" that only have ground truth 2D annotations. However, relying solely on the reprojection loss leaves the model under-constrained. To address this challenge, the authors introduce an adversarial component trained to discern whether a human body parameter is real or not. This adversarial network leverages a large database of 3D human meshes. It directly infers 3D pose and shape parameters from image pixels, and operates in real-time, given a bounding box containing the person.

Pavlakos et al. (2018b) develop PosePrior, a network trained to accept heatmaps as input, generating pose parameters for the SMPL model. They also employ a separate network named ShapePrior to infer shape parameters from the silhouette. The authors demonstrate that these parameters can be reliably predicted solely from BigPicture2D keypoints and masks, which are typical outputs of generic 2D HPE through CNNs. Subsequently, the resulting 3D pose is compared with annotated keypoints and masks. The 3D mesh is explicitly optimized for the surface using a 3D per-vertex loss. The approach outperforms previous baselines on this task, offering an attractive solution for directly predicting 3D shape from a single color image. Through utilizing semantic segmentation for body parts and considering the body constraints of SMPL, Omran et al. (2018) proposed the Neural Body Fitting (NBF) model. It integrates a statistical body model within a CNN by partitioning the image into 12 semantic body parts. Subsequently, they encode the probability maps of these semantic parts into SMPL parameters. This combination leverages reliable bottom-up semantic body parts segmentation and robust top-down body model constraints. Estimating 3D human poses from images often results in implausible body configurations—bodies that lean, float, or even penetrate the floor. Existing methods tend to ignore the fact that bodies are typically supported by the scene. Tripathi et al. (2023) developed IPMAN, utilizing Intuitive-Physics (IP) terms inferred from a 3D SMPL body's interaction with the environment. Inspired by biomechanics, they deduced the pressure heatmap, Center of Pressure (CoP), and Center of Mass (CoM) to predict a 3D body from a color image in a *stable* configuration. It encourages plausible floor contact and ensures that the CoP and CoM overlap. The IP terms are user-friendly, easy to implement, computationally efficient, differentiable, and integrable to existing optimization and regression methods. IPMAN produces more realistic results compared to state-of-the-art methods, especially for static poses, without compromising dynamic ones. Code and data are available at: https://ipman.is.tue.mpg.de/

Multi-view images. Various approaches have been developed to integrate information from multiple points of view. In a multi-view setup, several cameras are positioned around the subject. These cameras capture images or videos of the same scene from different angles. Each camera provides a different view of the subject. When multiple cameras observe the same scene, they create overlapping views. By analyzing the geometry of these overlapping views, we can triangulate the 3D position of points in the scene. Triangulation involves finding the intersection point of rays projected from different camera viewpoints onto the same 3D point.

In a recent study by Qiu et al. (2019), multi-view images are fed into a CNN model to integrate information from different views into the current one. A cross-view fusion scheme is introduced into a CNN to jointly estimate 2D poses for multiple views. The second step focuses on recovering 3D poses from the multi-view 2D poses. The authors introduced a recursive 3DPS (3D Pictorial Structure) to enhance 3D pose optimization, gradually minimizing quantization errors for improved results with reasonable computational cost.

Training accurate 3D human pose estimators typically requires a large amount of 3D ground-truth data, which can be costly to collect. Various weakly or self-supervised pose estimation methods have been proposed due to the lack of 3D data. However, these methods often require additional supervision or camera parameters in multi-view settings (triangulation). EpipolarPose (Kocabas et al., 2019) employs Epipolar Geometry to reconstruct the 3D pose from 2D poses, utilizing this as a supervisory signal to train the 3D pose estimation model. Nowadays, Epipolar Geometry is a fundamental concept in 3D computer vision that plays a crucial role in stereo vision. Consider two cameras observing the same 3D scene from distinct positions. Each camera captures a 2D image of the 3D world. Epipolar geometry describes the geometric relations between the 3D points and their projections onto the 2D images. These relations lead to constraints between corresponding image points. Each camera's optical center projects onto a distinct point in the other camera's image plane. These points are called epipoles or epipolar points. Given a 3D point of interest, the line connecting the optical center to the point in one camera corresponds to a point in the other camera. This line in the other camera's image is called an epipolar line. These lines help estimate the depth of 3D points using triangulation. A common problem with HPE methods that focus on silhouettes is that they encounter difficulties when dealing with individuals wearing loose clothing or part of the body are obscured. To overcome these limits, Zhang et al. (2021) introduced AdaFuse, an adaptive fusion method improving features in obstructed views by leveraging unobstructed view data, achieved through effective point-to-point correspondence determination using heatmap sparsity. It learns adaptive fusion weights for each camera view to prevent "bad" views from corrupting good features, training alongside a pose estimation network and easily applicable to new camera setups without additional adaptation.

In their recent work, Wandt et al. (2021) proposed a novel method for self-supervised 3D pose estimation using unlabeled multi-view data. Their approach involves leveraging multi-view consistency constraints to separate 2D pose into underlying 3D pose and camera rotation. Notably, their method does not require calibrated cameras and can learn from moving cameras as well as static setups. By integrating information across views and training samples with unbiased reconstruction objectives, they achieved successful results.

TesseTrack by Reddy et al. (2021) concurrently reconstructs 3D body joint movements of multiple individuals in a single learnable framework, using a unique spatio-temporal formulation in a voxelized feature space from one or multiple camera views. Voxels refers to a representation of a three-dimensional (3D) object or scene that has been converted into a discrete grid of small cubic elements. Short-term individual representations are generated by a 4D CNN, linked across time via a differentiable matcher, and merged into 3D poses, enhancing robustness to camera view variations and achieving strong performance even with a single view during inference, unlike previous piecewise methods.

Recent approaches proposed by Chun et al. (2023a) proposed Learnable human Mesh Triangulation (LMT), a technique that initially predicts the positions of mesh vertices using a CNN model based on input images. Then, they obtain SMPL parameters by training the SMPL model on predicted mesh vertices. The predicted mesh vertices offer ample data for discerning joint rotation and shape, making them simpler to learn compared to direct SMPL parameters. This approach attained top-tier performance, setting a new state-of-the-art on the MPI-INF-3DHP dataset (Mehta et al., 2017a). Chun et al. (2023b) demonstrated that utilizing an autoencoder to learn vertex heatmaps enhances the effectiveness of these methods. Their Vertex Heatmap Autoencoder (VHA) captures the manifold of feasible human meshes by learning latent codes from AMASS (Mahmood et al., 2019), a vast motion capture dataset. The Body Code Predictor (BCP) leverages the body prior learned from VHA to reconstruct human meshes from multi-view images using latent code-based guidance and the transfer of pretrained weights. BCP achieves state-of-the-art performance in human mesh reconstruction by providing valuable insights for further research investigation into representation learning, latent codes, and mesh reconstruction.

Sequence of monocular images. Estimating the 3D pose of a human from a series of single-lens images is challenging due to inherent depth ambiguity, making it an ill-defined problem. Many approaches address this issue by taking an image sequence as input. However, this can lead to multiple images of the same person in successive frames. Despite this, the bone length and shape of an individual remain constant, and their movements often follow regular patterns, helping to alleviate ambiguities in the estimation process.

Hossain and Little (2018) leveraged temporal data from a series of 2D joint positions to predict a series of 3D poses. This approach involves constructing a sequence-to-sequence network, which consists of layer-normalized LSTM units, while the authors incorporated shortcut connections linking the input to the output on the decoder side and enforce a temporal smoothness constraint during the training process. Hu and Ahuja (2021) introduced the Hierarchical Dance Video Recognition framework (HDVR), which estimates 2D pose sequences, tracks multiple dancers, and simultaneously predicts 3D poses and imaging parameters without ground truth for 3D poses. Unlike single-person methods, HDVR handles multiple dancers and occlusions, extracting body movements to identify dance genres in a hierarchical representation. To address noise and correspondence ambiguities, they enforce spatial and temporal motion smoothness and photometric continuity, using an LSTM network to recognize dance genres from extracted 3D movement subsequences.

Various network architectures have been investigated to address the challenge of 3D HPE and temporal information. Among the most relevant ones, Dabral et al. (2018) introduce the Temporal PoseNet, a straightforward two-layer Fully Connected Network (FCN) employing Rectified Linear Units (ReLUs). Temporal PoseNet aims to predict 3D human poses by leveraging both temporal information and 2D joint coordinates obtained from video frames. This model accepts a predefined set of neighboring poses as input and generates the corresponding 3D one, demonstrating the capability to grasp intricate structural and motion cues. In response to the challenge of occlusion, Cheng et al. (2019) suggest an occlusion-aware 2D temporal CNN. This model utilizes an incomplete sequence of 2D poses as input, aiming to mitigate the impact of inaccuracies in estimating occluded joints. Specifically, they advocate employing a "Cylinder Man Model" to create 2D-3D pose pairs with occlusion labels by inserting cylinders as body parts approximations and masking them. These data are then used to train the 3D Temporal Convolutional Network and enhance the consistency of occluded joints. Cai et al. (2019) depict a 2D pose sequence in the form of a graph and devise a Local-to-Global network, which is capable of learning multi-scale graph features via successive graph pooling and upsampling layers, for predicting the corresponding 3D pose sequence. Indeed, the network is capable of capturing features at multiple scales and acquiring a temporal constraint for the pose sequence.

When implementing these approaches, various considerations are taken into account, including ensuring the temporal consistency of both pose and shape, as well as employing diverse pose reconstruction procedures. Crucially, in contrast to methods based on a single time point, maintaining the temporal consistency of human pose or shape is emphasized, achieved through penalizing the relevant parameters.

Xu et al. (2018) addresses the reconstruction of articulated human skeleton motion and medium-scale non-rigid surface deformations in general scenes by proposing the MonoPerfCap framework. To overcome the challenges of monocular video reconstruction, they use a novel approach leveraging sparse 2D and 3D human pose detections from CNNs, employing batch-based pose estimation to handle occlusions and depth ambiguity, and proposing surface geometry refinement based on automatically extracted silhouettes for medium-scale non-rigid alignment. Given the shape representation, they estimate the deformation of the person for each frame in the input video, such that the deformed template closely matches the input frame. The resulting algorithm allows them to generate a temporally coherent surface representation of the person's full body performance.

Li et al. (2019a) adopt a two-stage framework to utilize unannotated monocular videos. The authors aimed to enhance single-frame 3D human pose estimation using monocular videos. The challenge lies in obtaining accurate 3D annotations, which are often time-consuming and error-prone. In the first stage, initial predictions are obtained from a pose estimation network (baseline model) trained on only a few annotated videos. Reliable estimations produced by the baseline model are fixed. These reliable estimations serve as anchors to automatically collect annotations across the entire video. This process addresses the 3D trajectory completion problem, i.e. reconstructing complete 3D trajectories from sparse or partial observations. In the second step, the baseline model is then further trained using the collected annotations to learn new poses. Notably, this method does not rely on multi-view imagery or explicit 2D keypoint annotations.

Li et al. (2021) presented the Lifting Transformer, a Transformer-based model for 3D HPE, using a Vanilla Transformer Encoder (VTE) to capture long-range dependencies in 2D pose sequences. They replaced VTE's fully-connected layers with strided convolutions to create the Strided Transformer Encoder (STE), which reduces computation cost, aggregates information globally and locally, and employs a full-to-single supervision scheme with temporal smoothness constraints at both full sequence and single target frame scales. A strided convolution is a type of convolution where the filter (also known as the kernel) skips some pixels as it moves over the input data. Unlike regular convolutions that slide step-by-step, strided convolutions take larger steps, effectively downsampling the input. The stride refers to how many pixels the filter moves horizontally and vertically during each step. Strided convolutions encourages the network to focus on the most discriminative features while ignoring redundant information, and it reduces the spatial dimensions of feature maps.

Yuan et al. (2021b) introduced SimPoE, a simulation-based method for 3D HPE, integrating image-based kinematic inference and physics-based dynamics modeling. It combines the strengths of both approaches to achieve accurate and physically plausible pose estimation. SimPoE employs a reinforcement learning system policy to jointly use a strategy based on a kinematic pose refinement unit with 2D keypoints and a dynamics-based control generation. The model takes the current-frame pose estimation and the next image frame as input. Then it controls a physically-simulated character to output the next-frame pose estimate. Additionally, the authors propose a meta-control mechanism for dynamically adjusting the character's dynamics parameters based on its state, enhancing pose estimate accuracy.

Zheng et al. (2021) introduced PoseFormer, a Transformer-based approach for 3D HPE in videos, deviating from traditional convolutional architectures. Inspired by Vision Transformers, they designed a spatial–temporal transformer structure that comprehensively models human joint relations within frames and captures temporal correlations across frames. As a result, the model accurately predicts the 3D human pose for the center frame. Unlike traditional heatmap-based methods, PoseFormer directly predicts joint positions and types. It achieves this using learnable object queries within the standard transformer architecture. PoseFormer showcases the potential of Transformers for 3D human pose estimation, emphasizing spatial–temporal modeling and heatmap-free predictions.

Hu et al. (2021) introduced a novel approach representing the human skeleton as a Directed Graph with joints as nodes and bones as edges, emphasizing hierarchical relations. They developed spatial–temporal directed Graph Convolution and spatial–temporal Conditional Directed Graph Convolution integrated into a U-shaped network for 3D HPE in monocular videos.

Li et al. (2022b) introduced MHFormer, a model capturing spatio-temporal representations of multiple pose hypotheses. MHFormer explicitly models multiple plausible pose hypotheses. It decomposes the task into three stages: (i) it generates diverse initial hypotheses; (ii) it merges and splits hypotheses; (iii) the method enhances the overall representation, resulting in a more accurate synthesized 3D pose. MHFormer is based on a cascaded Transformer-based architecture able to generate features at different depths of the latent space.

Zhang et al. (2022) introduced MixSTE (Mixed Spatio-Temporal Encoder), featuring a Temporal Transformer block to individually capture the temporal motion of each joint and a Spatial Transformer block for learning inter-joint spatial correlation. These blocks are employed alternately to enhance spatio-temporal feature encoding. Furthermore, the network output extends from the central frame to encompass the entire input video frames, enhancing coherence between input and output sequences.

Gholami et al. (2022) introduced AdaptPose, a comprehensive system that creates artificial 3D human movements from a reference dataset and employs them to refine a 3D pose estimation model. It addresses the challenge of cross-dataset generalization in 3D human pose estimation models. When testing a pre-trained 3D pose estimator on a new dataset, there is often a significant performance drop. Gholami et al. statet that the characteristics of the training data need to be adapted to those of the new dataset, considering factors such as camera viewpoint, position, human actions, and body size. Utilizing an adversarial training approach, AdaptPose's generator produces a series of 3D poses and a camera orientation from a source 3D pose, projecting these poses into the new dataset perspective. AdaptPose proficiently learns to generate synthetic 3D poses from a target dataset without requiring explicit 3D labels or camera data, relying solely on 2D pose information during training.

Zhu et al. (2023) introduced a pretraining phase where a motion encoder learns to reconstruct the underlying 3D motion from incomplete 2D observations, incorporating geometric, kinematic, and physical insights about human movement. MotionBERT is a groundbreaking framework that unifies the learning of human motion representations from large-scale and heterogeneous data resources. The pre-train phase is grounded over a Motion Encoder. The Motion Encoder captures long-range spatio-temporal relationships among skeletal joints comprehensively and adaptively. When trained from scratch, MotionBERT achieves the lowest 3D pose estimation error reported so far on the Human3.6M dataset (Ionescu et al., 2014). Remarkably, its performance on different HPE tasks are achieved by simply fine-tuning the pretrained motion encoder with a simple regression head (1-2 layers). The source code of the approach is available on GitHub: https://motionbert.github.io/.

Zhao et al. (2023) proposed PoseFormerV2 which exploits a compact representation of lengthy skeleton sequences in the frequency domain to efficiently scale up the receptive field and boost robustness to noisy 2D joint detection. With minimum modifications to PoseFormer, the proposed method effectively fuses features both in the time domain and frequency domain, enjoying a better speed-accuracy trade-off than its precursor.

Mehraban et al. (2024) proposed MotionAGFormer which leverages GCNFormer to capture intricate local joint relationships, and combines it with Transformer, which effectively captures global joint interdependencies. This fusion enhances the model's ability to comprehend the inherent 3D structure within input 2D sequences. Additionally, MotionAGFormer offers various adaptable variants, allowing the selection of an optimal balance between speed and accuracy.

Peng et al. (2024) proposes Kinematics and Trajectory Prior Knowledge-Enhanced Transformer (KTPFormer), which leverages the anatomical structure of the human body and motion trajectory information to enhance the learning of global dependencies and features in multi-head self-attention. KTPFormer is based on two prior attention modules: Kinematics Prior Attention (KPA) and Trajectory Prior Attention (TPA). KPA models the kinematic relationships within the human body by constructing a kinematic topology, while TPA builds a trajectory topology to capture joint motion trajectories across frames.

Zhang et al. (2024) introduces the Adaptive Pose Pooling (APP) module, designed to extract temporal and spatial information from multi-frame 2D human pose data and feature maps. The core idea of this module is to use feature maps generated by a 2D pose detector, along with multi-frame lifting models' calculated pose features, while prioritizing features close to the 2D poses as more relevant. The APP module consists of three sub-modules: Pose-Aware Offsets Generation (PAOG), which leverages 2D human poses as indices to learn offsets during training; Pose-Aware Sampling (PAS), which adaptively extracts features from feature maps generated by the 2D pose detector; and Spatial Temporal Information Fusion (STIF), which fusing image and temporal features to facilitate information interaction between the features sampled by PAS and the hidden features of the lifting model. The APP module further enhances performance when combined with state-of-the-art models like MotionBERT and MotionAGFormer.

Sequence of multi-view images. Some recent works belong to this classification and are presented individually below. Traditional approaches to estimating 3D human pose from multiple views have been extensively researched and outperform single-view methods. Nevertheless, these methods necessitate costly dense camera setups and controlled studio environments.

Rhodin et al. (2016) introduced an algorithm for automated creation of a detailed actor model for animation, predicting motion from multi-view video input. The method incorporates a novel image formation model, utilizes Gaussian Density field for 3D body shape reconstruction, and employs a unique Edge-based Alignment Energy inspired by volume ray casting to infer the visibility of the background. Based on their proposed statistical body model, the approach reconstructs a personalized kinematic skeleton, a volumetric Gaussian density representation with appearance modeling, a surface mesh, and the time-varying poses of an actor.

Huang et al. (2017) introduced a fully automated approach that, when provided with multi-view videos, predicts 3D human pose and body shape. Building upon the SMPLify method (Bogo et al., 2016), they enhance it in several ways. Initially, a 3D human body model with 2D features identified in multi-view images is aligned. Subsequently, a CNN method is applied to segment the individual in each image, and the 3D body model is fitted to the contours, enhancing overall accuracy. Additionally, they incorporate a generic and robust Discrete Cosine Transform (DCT) (Ahmed et al., 1974) temporal prior to address the occasional left and right side swapping issue introduced by the 2D pose estimator, suppressing noise while preserving video details.

Trumble et al. (2018) proposed a technique that concurrently gauges 3D human pose and body shape through a limited array of wide-baseline camera perspectives. It utilizes a Symmetric Convolutional Autoencoder with a dual loss to acquire a latent representation encoding skeletal joint positions and a profound representation of volumetric body shape. The method accomplishes real-time (25 fps) inference, enlarging volumetric data by 4× while preserving precise 3D joint position estimation. This renders it applicable for meticulous monitoring of human body shape and pose in situations demanding unobtrusive observation of human behavior.

Table 3 reports the key differences of 3D HPE methodologies, highlighting their pros and cons.

3.2.2. 3D multi-person pose estimation

Multi-person 3D pose estimation is a challenging task with several complexities that make it more intricate than single-person 3D pose estimation. The state space is significantly larger, and occlusion issues and cross-view ambiguities can arise when identities are unknown in advance. This section delves into the complexities of multi-person 3D pose estimation and highlights the differences from the single-person scenario. The challenges include distinguishing joint points and body parts that are nearby or obscured by occlusion, localizing root joints with assumptions that differ from single-person methods, and ensuring efficiency despite the increased number of persons to detect body bounding boxes or joints accurately. Despite these challenges, existing multi-person approaches employ two-stage strategies that have proven effective. The top-down method uses detectors to initially locate person instances and subsequently localize individual joints, while the bottom-up approach first identifies all body joints and then assigns them to their corresponding individuals. By using these strategies, multi-person 3D pose estimation can be achieved with high accuracy and speed.

Single monocular image. Rogez et al. (2020) proposed an end-to-end architecture for 2D and 3D HPE in wild environment images, called LCR-Net, that relies on generating and scoring multiple pose candidates for each image. This approach allows the simultaneous prediction of poses for multiple individuals without requiring an approximate localization of humans as an initialization step. LCR-Net comprises three main components: a pose proposal generator suggesting candidate poses, a classifier scoring these proposals, and a regressor refining poses in 2D and 3D. All stages share CNN layers and are jointly trained. The final pose estimation involves integrating neighboring pose hypotheses, demonstrating improvement over standard Non-Maximum Suppression algorithms (Papandreou et al., 2017).

Moon et al. (2019) introduced an innovative and versatile framework designed for 3D multi-person PE from a single RGB image. Their approach encompasses human detection, 3D human root localization, and root-relative 3D single-person pose estimation models. Its flexibility and ease of use arise from the adaptability to integrate any existing human detection and 3D single-person PE models into their framework.

Mehta et al. (2018) proposed a method that utilizes innovative Occlusion-Robust Pose-Maps (ORPM) to enable robust body pose inference even in the presence of significant partial occlusions from other individuals and objects. ORPM generates a fixed number of maps encoding 3D joint locations for all individuals in the scene, allowing 3D pose deduction without explicit bounding box predictions through body part associations. For training, the MuCo-3DHP dataset was introduced. It can be considered as the first large-scale dataset with real images depicting complex multi-person interactions and occlusions, created by composing individual images with ground truth from multi-view performance capture.

Nie et al. (2019) challenged the inefficiencies of two-stage methods in multi-person pose estimation by introducing the Single-stage multi-person Pose Machine (SPM). SPM used the Structured Pose Representation (SPR) to unify person instances and body joint positions, by predicting structured poses for multiple persons in a single stage. SPR introduced in its architecture root joints and hierarchical representations for efficient long-range displacement predictions, demonstrating SPM's versatility in multi-person and 3D pose estimation.

Table 3
Comparison of 3D HPE methods.

Method	Key Differences	Pros	Cons
Direct Prediction of 3D Poses	Directly maps image to 3D coordinates using a single deep network	Simple pipeline; fewer intermediate steps	Requires large 3D datasets; sensitive to depth ambiguities
Lifting from 2D Poses	Uses 2D pose estimation as an intermediate step before lifting to 3D	Reduces need for 3D data; can use existing 2D pose models	Depends on 2D pose quality; struggles with depth ambiguities
SMPL Models	Parametric body model for generating 3D meshes with shape and pose parameters	Detailed and realistic body meshes; captures surface deformations	Computationally expensive; requires complex fitting
Multi-View Images	Leverages multiple synchronized cameras to triangulate and reconstruct 3D poses	High precision; robust to occlusions and depth ambiguities	Requires multi-camera setup and calibration; limited to controlled environments
Sequence of Monocular Images	Uses temporal context from a sequence of monocular images to estimate 3D poses over time	Leverages motion patterns; handles occlusions better	Needs continuous sequences; high memory usage
Sequence of Multi-View Images	Combines information from multiple cameras over time, using both temporal and spatial context	Achieves very high accuracy; resolves occlusions and ambiguous poses; strong temporal and spatial cues	Requires large-scale multi-camera setups and synchronization; high computational and storage costs

Multi-view images. The main challenge in multi-person 3D pose estimation from multi-view images is the need to establish cross-view correspondences among 2D pose predictions. This is difficult because the 2D poses may be incomplete or have low confidence levels. Therefore, it is important to come up with new strategies to overcome this bottleneck, and many efforts have already been made to address this issue.

Belagiannis et al. (2014a) have developed an approach to tackle the challenges of large state space and partial occlusions in human pose estimation (HPE). They achieve this by generating a smaller state space through the triangulation of corresponding body joints identified by part detectors across pairs of camera views. To deal with uncertainties arising from incorrectly identified or mixed body parts of multiple individuals post-triangulation, as well as those resulting from false positive body part detections, they introduce the pioneering 3D Pictorial Structures (3DPS) model. This model is designed to deduce 3D configurations of the human body based on the reduced state space. It is noteworthy that the 3DPS model is versatile and can be used in both single and multiple HPE scenarios.

Dong et al. (2019) introduced a new approach to multi-view 3D pose estimation that allows for the rapid and accurate recovery of 3D poses for a crowd using only a few cameras. They improved upon prior methods that relied on 3DPS by applying a multi-way matching algorithm to cluster detected 2D poses. This clustering effectively reduced the state space of the 3DPS model, which led to significant improvements in both efficiency and robustness. The authors also demonstrated that accurate reconstruction of 3D poses can be achieved from clustered 2D poses through triangulation, without relying on the 3DPS model. This highlights the effectiveness of the proposed multi-way matching algorithm, which combines geometric and appearance cues with the cycle-consistency constraint to match 2D poses across multiple views.

Rhodin et al. (2019) proposed a self-supervised multi-view learning approach to train a network for generating a hierarchical scene representation. The proposed method is primarily designed for 3D human pose capture, but it is flexible enough to be used in other reconstruction tasks. The scene representation consists of three levels of abstraction: spatial layout (bounding box and relative depth), instance segmentation (masks), and body representation (latent vectors encoding appearance and pose). The proposed method requires minimal annotated data to train a secondary network that maps the scene representation to a complete 3D pose. The trained network can then be used without predefined locations for individuals, and it can efficiently calculate their poses in parallel, even in overlapping scenarios.

Dong et al. (2021) proposed a pipeline that uses a coarse-to-fine strategy to convert imprecise 2D observations from multiple camera views into 3D space. The observations are then associated with individual instances using a confidence-aware majority voting technique. Pose estimates are obtained through a novel optimization scheme that connects high-confidence multi-view 2D observations and 3D joint candidates. The inclusion of a statistical parametric body model, such as SMPL, as a regularization prior, enhances accuracy by refining 3D poses and SMPL parameters alternately. This correction of implausible estimates also improves SMPL estimations. The method shows accuracy, generalizability across diverse data sources, and robustness to noisy 2D detections. This is achieved by effectively isolating the final 3D pose from interperson dependencies.

Sequence of monocular images. Zanfir et al. (2018) introduced a monocular model that can estimate multiple people's 2D and 3D poses and shapes in complex scenes. The model uses feedforward predictors for initialization and semantic fitting to refine the shape of individuals based on the scene layout. Semantic fitting refers to the process of aligning data (such as images, or other representations) with semantic structures or models. It aims to find the best correspondence between data and underlying semantic concepts. It considers ground plane and volume occupancy constraints, along with temporal priors, to ensure consistent and plausible results. The optimization process involves a unified representation of multiple people in both spatial and temporal dimensions. The study covers datasets with single-person and multiple-person scenarios and highlights the importance of incorporating diverse constraints. Qualitative assessments demonstrate that the method produces 3D reconstructions with precise image alignment and satisfactory perceptual quality. It is applicable to monocular images and videos captured in complex scenes featuring multiple individuals, substantial occlusion, and challenging backgrounds.

Sequence of multi-view images. Estimating the 3D poses of multiple individuals from various camera perspectives in uncontrolled environments poses a significant challenge. The task involves matching each person across different views and subsequently estimating their body poses. Moreover, the body poses of individuals exhibit consistent changes over time. To tackle these difficulties, Belagiannis et al. (2014b) proposed a 3D Pictorial Structures model (3DPS) specifically designed for multiple HPE from multiple camera views, emphasizing temporal consistency. The model extended the 3DPS framework by introducing the concept of temporal coherence in the inferred body poses. This temporal consistency is achieved through the incorporation of multi-view human tracking. By identifying each individual before the inference stage, they effectively reduce the state space, positively impacting the overall performance of the model.

3.3. Non-camera-based human pose estimation

Camera-based HPE solutions face several technical and social limitations. Technical challenges such as the diversity of clothing, backgrounds, lighting, and occlusion, coupled with the lack of distance information, hinder the accuracy of camera-based HPE. The estimated distance information from camera parameters also has a significant gap from the actual situation. On the other hand, non-camera-based approaches, such as using radar signals, are not human-interpretable, making labeling them a difficult and time-consuming task. Human workers can label RGB datasets, but it is infeasible for humans to annotate radar signals with keypoints. In such non-camera-based approaches, the ground truth is often directly obtained from the camera-based approach, where frames are acquired simultaneously with non-camera-based methods. The challenging limitations of this family of approaches are mainly:

- Radar data is different from RGB images because it can only receive the amplitude and phase of electromagnetic waves. This means that the received amplitude is a one-dimensional summary of the 3D space, which makes it difficult to estimate human pose accurately.
- Unlike outdoor environments, where radar signals propagate relatively directly, indoor settings often exhibit a significant amount of multipath interference. This occurs when radar signals bounce off multiple objects in the environment, creating multiple paths to the receiver. As a result, the received signal becomes a complex combination of these paths, making it challenging to isolate the signal reflected by the human body and accurately estimate their pose.
- Radar-based HPE systems approaches employ multiple transmitting and receiving antennas to improve the accuracy of pose estimation. Indeed, the cost of this solution is not always affordable.

Among non-camera-based approaches, radar-based ones are the most discussed in the scientific literature. For this reason, it is worth summarizing at least the most common categories:

- WiFi-based: The researchers investigated the possibility of exploiting Channel State Information (CSI) from WiFi devices to estimate human poses. Wang et al. (2019d,c) pioneered a deep learning approach that utilizes 2D image annotations to directly map received 1D WiFi signals to human poses. However, these methods are limited to specific viewpoints or require subjects to remain stationary, hindering their real-world applicability. Guo et al. (2020) proposed Wi-Pose, which leverages synchronized video frames to guide pose estimation from CSI. Their neural network captures fine-grained human skeleton images, but performance suffers when applied to unseen individuals. Wang et al. (2021a) further refines this approach by extracting pose-specific features from CSI images and converting them to keypoint coordinates. However, generalization to new subjects remains challenging. DINN (Zhou et al., 2021) addresses this issue by employing a domain-independent deep learning network to extract pose features and construct fine-grained human pose images. Despite these advancements, commercial WiFi systems need multiple antennas to properly capture human pose, limiting practical deployment. Moreover, lower frequency WiFi bands (2.4 GHz and 5 GHz) with standard bandwidths (20 MHz and 40 MHz) provide insufficient resolution for detailed pose estimation. Temporal fluctuations due to multipath propagation and interference from the surrounding environment further hinder the system's effectiveness, restricting its range of applications.
- RFID-Based: Radio Frequency Identification (RFID) technology has emerged as a promising tool for various applications, including indoor localization, vital sign monitoring, and drone localization and navigation. RFID tags, due to their small size and convenient form factor, can be seamlessly integrated into wearable devices. The RFID-Pose system by Yang et al. (2021b) paved the way for 3D HPE using RFID technology, demonstrating the ability to simultaneously track multiple human joints in real-time. The Meta-Pose system (Yang et al., 2021c) further enhanced this approach by incorporating meta-learning and few-shot fine-tuning, enabling the system to adapt effectively to new environments. To address the challenge of generalization across different subjects, a novel deep learning model, a Cycle Consistent Adversarial Network (Yang et al., 2022), was developed. However, the low data rate characteristic of RFID systems poses a significant challenge in generating joint confidence maps for all joints, similar to other RF-based systems.
- LiDAR-based: LiDAR (Light Detection and Ranging or Laser Imaging Detection and Ranging) is a remote sensing instrument that measures the distance to objects or surfaces using laser pulses. Unlike cameras and WiFi, which rely on indirect depth estimation, LiDAR provides accurate 3D spatial information. This precise depth information is crucial for accurate HPE, which has been hindered by the limitations of traditional camera-based methods. To address this issue, Fürst et al. (2020) developed an end-to-end architecture that combines RGB camera images with LiDAR data to achieve exceptional 3D HPE accuracy. However, the high cost of LiDAR sensors hinders their widespread adoption in HPE systems.
- Radar-based: Researchers have been investigating contact-free sensing methods to reconstruct human posture in order to address the drawbacks of body-mounted sensor- and vision-based approaches. The first category uses very high frequencies (e.g., millimeter-wave - mmWave, or terahertz) (Sengupta et al., 2020, 2019; Li et al., 2020b; Shi et al., 2022; Wang et al., 2020; Sengupta and Cao, 2023; Cui and Dahnoun, 2022). The shorter wavelengths of mmWave signals enable more precise posture capture but do not penetrate walls or furniture. The first method (Sengupta et al., 2020, 2019; Sengupta and Cao, 2023) that utilizes mmWave radar reflection signals to determine the real-world position of distinct body joints, employing mmWave radars and CNNs. Shi et al. (2022) design a domain discriminator that eliminates user-specific traits embedded in the mmWave signals to achieve robust skeleton reconstruction across users with a reasonable amount of training effort. The second category employs

lower frequencies, around a few GHz, which offers several advantages: it can traverse walls and obstructions, function during the day and night, and is not human-interpretable, hence more privacy-preserving. MIT researchers introduced a neural network system (Zhao et al., 2018a,b; Li et al., 2022a), that analyzes radio signals to estimate 2D human poses and dynamic 3D human meshes. Song et al. (2021a) developed a new through-wall 3D pose reconstruction framework for concealed target detection using UWB MIMO radar and 3D CNN. Zheng et al. (2022) proposed a novel cross-modal CNN-based human postural reconstruction method for TWRI (Through-the-wall radar imaging).

There are several works that use Millimeter-wave (mmWave) radars, as mentioned in Sengupta and Cao (2023). The motivation behind using this technology is that it is not affected by external lighting or weather conditions. This is because it uses its own RF signals to radiate the target and captures the target's silhouette via a sparse 3D point cloud (PCL) representation. It does not capture facial features of the users, thus ensuring privacy. Additionally, these radars are low-cost, low-power, and compact, making them a practical and non-invasive monitoring solution. This technology is suitable for applications with privacy concerns such as surveillance systems in patient rooms. More details about works grounded on this technology will follow. In 2019, mm-Pose (Sengupta et al., 2019) was developed to estimate skeletal pose in 3D space using Kinect (Zhang, 2012; Tölgyessy et al., 2021) as ground truth, based on 25 joints. Their first approach employed a forked-CNN-based deep learning architecture. However, their second work, mmPose-NLP (Sengupta and Cao, 2023), implemented a model with an architecture similar to a seq2seq model (Sutskever et al., 2014). This new approach yielded a performance improvement of 2 points on the MAE metric comparing their best result to their approach. In 2020, a comparable method to mm-Pose was developed (Li et al., 2020b), which also employs mmWave radar with a CNN. However, instead of utilizing 3D point cloud data, this method first generates heatmaps from raw vertical and horizontal dimension data. The CNN is then applied to convert these two-dimensional heatmaps into human pose estimates. In 2023, a new HPE benchmark called HuPR was introduced by Lee et al. (2023). This benchmark distinguishes itself from previous works by incorporating velocity information and employing a novel fusion technique to enhance accuracy. Additionally, HuPR includes a 2D pose confidence refinement network based on Graph Convolutional Network to refine the confidence in the output pose heatmaps.

The first attempt to use transformers architecture in radar-based pose estimation is 3D-TransPOSE (Kim et al., 2023). This method utilizes an *IR-UWB MIMO radar*, which utilizes low-frequency RF signals that can penetrate through moderate-thickness concrete walls. This makes 3D-TransPOSE more suitable for through-wall radar applications than previous approaches. To evaluate the performance of their transformer-based approach, the authors conducted an ablation study using a CNN architecture on the same dataset. The results confirmed that the transformer outperformed the CNN, demonstrating the effectiveness of transformers for radar-based pose estimation.

The approach proposed by Cao et al. (2023) explores how mmWave signals pose a challenge to developing a consistent feature representation. The signals reflected from different subjects can vary significantly even when they are performing the same poses. To address this challenge, the study proposes a framework that combines domain adversarial learning and disentanglement representation learning. It aims to isolate task-specific features. Specifically, a two-stage feature selection approach is developed for radar-based HPE. This approach effectively extracts pose-specific features while diminishing the influence of irrelevant components that could compromise accurate human skeleton reconstruction. The first stage employs an adversarial autoencoder module to address inter-person discrepancies. Subsequently, a feature disentanglement module separates pose-specific factors from irrelevant ones with minimal dependency. The resulting selected feature demonstrates both task consistency and discriminability, contributing to enhanced human skeleton reconstruction accuracy. Extensive evaluations in-domain and cross-domain own datasets corroborate the effectiveness of the proposed method.

4. Frameworks, tools and libraries

In this section, our aim is to provide a comprehensive overview and guidance on the prevalent libraries, frameworks, and tools for implementing the methodologies described in Section 3. Through this summary, we intend to offer insights into the diverse array of resources available to researchers and practitioners. Additionally, we strive to enhance the accessibility and applicability of the discussed techniques by providing links to code repositories, detailed documentation, and pertinent usage examples.

Detectron2 is a library developed by Facebook AI Research that offers advanced detection and segmentation algorithms and serves as the successor of Detectron[1] and maskrcnn-benchmark[2] (Wu et al., 2019). It is used in various computer vision research projects and production applications at Facebook. Developed in PyTorch, Detectron2 is highly modular, flexible, and efficient. This makes it suitable for both research and production applications. Detectron2 is capable of handling a wide range of computer vision tasks, including object detection, instance segmentation, keypoint estimation, panoptic segmentation, human pose estimation and dense pose mapping. It includes several state-of-the-art algorithms and models such as DensePose, Mask R-CNN, RetinaNet, Cascade R-CNN, PointRend, DeepLab, ViTDet, and MViTv2. Its modular design allows researchers to customize components and extend the framework. It supports multi-GPU training and inference, making it suitable for large-scale applications. Detectron2 provides a rich Model Zoo with pre-trained models and baseline results, covering popular benchmarks like COCO, LVIS, and Cityscapes. Detectron2 models can be exported to TorchScript format or Caffe2 format for deployment, which allows seamless integration into production systems. Detectron2 has gained popularity in the computer vision community due to its performance and ease of use. Researchers actively contribute to its development and extend its capabilities constantly. Detectron2 is offered under the Apache-2.0 license, allowing for free use, modification, and distribution, even for closed-source derivative works.

Ultralytics YOLOv8[3] one of the latest of YOLO (You Only Look Once) (Redmon et al., 2016), a popular object detection and image segmentation model, was developed by Joseph Redmon and Ali Farhadi at the University of Washington. Launched in 2015, YOLO quickly gained popularity for its high speed and accuracy, and several versions have been subsequently released. Version 2 (Redmon and Farhadi, 2017) and version 3 (Redmon and Farhadi, 2018) have improved performance by modifying the architecture of the model. In version 4 (Bochkovskiy et al., 2020), a data augmentation, a new anchor-free detection head, and a new loss function have been introduced. In version 5 new features have been added, such as hyperparameter optimization, integrated experiment tracking and automatic export to popular export formats. Version 6 was open-source, and version 7 (Wang et al., 2023a,b) added additional tasks such as pose estimation on the COCO keypoints dataset. Version 8 builds on the success of previous versions, by introducing new features and improvements for enhanced performance, flexibility, and efficiency. It supports a full range of vision AI tasks, including detection, segmentation, pose estimation, tracking, and classification. The framework, available on GitHub,[4] allows training models on custom datasets using images or videos. It is offered under the AGPL-3.0 license, granting users freedom

[1] https://github.com/facebookresearch/Detectron/
[2] https://github.com/facebookresearch/maskrcnn-benchmark/
[3] https://docs.ultralytics.com/
[4] https://github.com/ultralytics/ultralytics

to run, study, share, and modify the software, but requiring derivative works to be open-source, if distributed.

OpenCV (Open Source Computer Vision) is a free, cross-platform library initially developed by Intel for real-time image processing. Its goal is to provide an easy-to-use infrastructure for building vision applications by offering over 500 functions. The OpenCV software has become a de-facto standard tool for all things related to Computer Vision. The OpenCV library contains over 2500 algorithms, extensive documentation, source code, and sample code for real-time computer vision. It also contains image processing capabilities, it support video stream processing, image stitching (combining multiple cameras), camera calibration, and diverse image pre-processing tasks. It also leverages NVIDIA CUDA and GPU acceleration since 2011, improving performance, and provides control over data movement between CPU and GPU memory. OpenCV provides solutions for real-time object detection, image segmentation, movement and gesture recognition, face recognition and augmented reality. This is offered under the Apache-2.0 license, allowing use, modify, distribute the software, and even create closed-source derivative works.

MediaPipe[5] is an open-source framework developed by Google for building pipelines to perform computer vision tasks on various sensory data, including video and audio. This provides support[6] for different tasks, i.e. face detection, face landmark detection, hand landmark detection, pose landmark detection, holistic landmarks detection, image segmentation and object detection, performed using different models, as BlazePose (Bazarevsky et al., 2020). MediaPipe supports hardware like GPUs, CPUs, or TPUs and development across platforms i.e. Android, iOS, desktop, edge, cloud, web, and IoT platforms. Similar to OpenCV and Detectron2, MediaPipe is offered under the Apache-2.0 license.

MMPose[7] is a Pytorch-based pose estimation open-source toolkit, a member of the OpenMMLab[8] Project. OpenMMLab team integrates open source, academic research, business application and system development, aiming at becoming a worldwide leader in open source platforms for computer vision and providing advanced R&D models and systems for the industry. It contains a rich set of algorithms for 2D multi-person human pose estimation, 2D hand pose estimation, 2D face landmark detection, 133 keypoint whole-body human pose estimation, fashion landmark detection and animal pose estimation as well as related components and modules. Several algorithms, techniques, datasets and backbones are supported. MMPose is offered under the Apache-2.0 license.

PaddleDetection[9] is an end-to-end object detection development kit based on PaddlePaddle. Providing over 30 model algorithm and over 300 pre-trained models, it covers object detection, instance segmentation, keypoint detection, multi-object tracking. In particular, PaddleDetection offers high- performance and light-weight industrial SOTA models on servers and mobile devices, champion solution and cutting-edge algorithm. PaddleDetection provides various data augmentation methods, configurable network components, loss functions and other advanced optimization and deployment schemes. In addition to running through the whole process of data processing, model development, training, compression and deployment, PaddlePaddle also provides rich cases and tutorials to accelerate the industrial application of algorithm. PaddleDetection is offered under the Apache-2.0 license.

ImgClsMob, a GitHub[10] repository that contains (re)implementations of various classification, segmentation, detection, and pose estimation models and scripts for training/evaluating/converting. This includes the usage of the frameworks as MXNet/Gluon, PyTorch, Chainer, Keras and Tensorflow with several pretrained models. This is provided under the MIT License, similar to the Apache-2.0 license, allowing the user to modify, freely distribute the software, including creating closed-source derivative works.

OpenPose (Cao et al., 2017), is a pioneering framework for real-time, multi-person, bottom-up pose estimation. It leverages DL techniques to accurately capture 2D poses by detecting key body joints and their interconnections. A distinctive feature of OpenPose is the use of Part Affinity Fields (PAFs) to model the associations between body parts, facilitating the precise estimation of body poses even in crowded or complex scenes with multiple individuals. The system is capable of handling diverse poses and occlusions, making it a robust tool for applications like human–computer interaction, virtual reality, and sport analysis. Its real-time capabilities and accuracy made OpenPose widely adopted in both research and practical settings, contributing significantly to advancements in the field of computer vision and PE. The current license for OpenPose restricts its use to academic or non-profit research purposes only.

EfficientPose (Groos et al., 2021) emerges as a noteworthy solution in the realm of single-person PE, emphasizing scalability and efficiency. Introduced as a technique to address the computational demands of the task, EfficientPose employs a scalable approach to ensure that accurate predictions can be made with minimal computational resources. The method optimizes the model architecture, harnessing the power of efficient NN that strike a balance between accuracy and computational efficiency. This design choice makes EfficientPose particularly well-suited for real-time applications and scenarios where single-person PE needs to be seamlessly integrated into resource-constrained environments. Its commitment to scalability extends beyond its efficient architecture. The method is engineered to handle diverse poses and variations in individual body articulations, ensuring reliable performance across different scenarios. Its scalability is a key attribute, allowing EfficientPose to be applied in a range of applications, including fitness tracking, human–computer interaction, and augmented reality, where the ability to scale pose estimation capabilities without compromising accuracy is paramount. This focus on scalability positions EfficientPose as a promising solution in the landscape of single-person pose estimation methodologies

AlphaPose[11] Fang et al. (2023) stands out as a state-of-the-art method in the realm of HPE, offering precise localization of body keypoints in images or videos. It employs a multi-stage framework that integrates both top-down and bottom-up strategies, ensuring robust performance in capturing intricate human poses. The method utilizes a carefully designed network architecture, combining spatial and temporal information to enhance the accuracy and temporal coherence of pose predictions. AlphaPose's versatility is evident in its ability to handle crowded scenes, occlusions, and variations in body articulations, making it applicable across diverse scenarios ranging from video surveillance to human–computer interaction. One distinctive feature of AlphaPose is its proficiency in real-time processing, making it suitable for applications where low-latency is critical. The method achieves this by leveraging efficient network architectures and optimization techniques, enabling swift and accurate pose estimation across frames. AlphaPose's success lies not only in its technical prowess, but also in its user-friendly design, empowering researchers and developers to integrate advanced HPE capabilities into a wide array of applications, including action recognition, healthcare monitoring, and sports analytics.

5 https://developers.google.com/mediapipe
6 https://github.com/google/mediapipe/tree/master/docs/solutions
7 https://github.com/open-mmlab/mmpose
8 https://openmmlab.com/
9 https://github.com/PaddlePaddle/PaddleDetection
10 https://github.com/osmr/imgclsmob
11 https://github.com/MVIG-SJTU/AlphaPose

5. Datasets for human pose estimation

The objective of this section is to describe the most prevalent datasets utilized in HPE. Understanding the landscape of datasets is paramount in HPE research as it lays the foundation for benchmarking algorithms, evaluating performance metrics, and fostering advancements in pose estimation techniques. By delving into the prominent datasets' characteristics, intricacies, and limitations, researchers and practitioners can gain valuable insights into the diverse scenarios and challenges encountered in real-world pose estimation applications. Moreover, a comprehensive exploration of the common datasets for HPE facilitates comparative analyses, fosters reproducibility in research findings, and drives innovation in algorithm development. By shedding light on the nuances of dataset composition, annotation protocols, and data diversity, the upcoming section aims to equip readers with a nuanced understanding of the underlying data sources that underpin advancements in human pose estimation technologies.

5.1. Popular datasets for 2D HPE

Initially established datasets for 2D HPE comprised images featuring relatively uncomplicated backgrounds. Nevertheless, these datasets prove inadequate for training Deep Learning models due to their limited image count. Contemporary Deep Learning approaches often rely on widely used datasets such as MPII, COCO, FLIC, LSP, Pose Track and AI Challenger. These datasets present a larger number of images set in more intricate scenes, rendering them more suitable for training Deep Learning models in 2D HPE. A summary of relevant statistics about 2D image datasets is reported in the Table 4.

5.1.1. Single-person dataset

Leeds sports pose (LSP) dataset. (Johnson and Everingham, 2010). It comprises 2000 images depicting full-body poses sourced from Flickr through downloads using eight sports-related tags, namely athletics, badminton, baseball, gymnastics, parkour, soccer, tennis, and volleyball. Each image is meticulously annotated, detailing up to 14 visible joint locations. Additionally, an extended version known as Leeds Sports Pose Extended (LSP-extended) is specifically assembled for training purposes, expanding the LSP dataset. This extended dataset encompasses 10,000 images gathered from Flickr searches utilizing the three most challenging tags: parkour, gymnastics, and athletics. It is possible to access the dataset at: https://datasets.activeloop.ai/docs/ml/datasets/lsp-dataset/

Penn action dataset. (Zhang et al., 2013). It is a challenging consumer video dataset where each frame includes action labels, 2D positions of keypoints, and their visibilities. The dataset, named Penn Action, is an unconstrained human action dataset sourced from YouTube, comprising 2326 video clips covering 15 types of actions. It includes 1258 training videos and 1068 test videos. Each person in the images is labeled with 13 keypoints, and both joint coordinates and visibility are provided. It is possible to access the dataset at: https://dreamdragon.github.io/PennAction/

Joint-annotated human motion DataBase (JHMDB) dataset. (Jhuang et al., 2013). It consists of 21 action categories involving a single-person action: brush hair, catch, clap, climb stairs, golf, jump, kick ball, pick, pour, pull-up, push, run, shoot ball, shoot bow, shoot gun, sit, stand, swing baseball, throw, walk, wave. Each category contains 36-55 video clips, with each clip comprising 15-40 frames. In total, there are 960 video clips with 31,838 annotated frames. Annotations are performed using a 2D puppet model (Zuffi et al., 2012), representing the human body with 10 body parts connected by 13 joints and two landmarks. A subset of the JHMDB dataset is also provided, called sub-JHMDB, containing 316 video clips with 12 action categories, with each person annotated with 15 joints. It is possible to access the dataset at: http://jhmdb.is.tue.mpg.de/

5.1.2. Multi-person dataset

Max Planck institute for informatics (MPII) human pose dataset (Andriluka et al., 2014). This dataset stands as a leading benchmark in the current landscape for evaluating articulated HPE, featuring comprehensive annotations. The dataset creation process involves downloading 3913 videos encompassing 491 distinct activities from YouTube, guided by a two-level hierarchy of human activities as proposed by Andriluka et al. (2014). Subsequently, manual selection of frames is undertaken, focusing on those containing different individuals within the video or the same person in notably different poses, resulting in a total of 24,920 frames containing over 40K people and 410 activities. The annotations, a product of both in-house workers and contributors on Amazon Mechanical Turk, encompass rich details, including the labeling of 16 body joints, such as shoulders, elbows, wrists, hips, knees, and ankles, the 3D viewpoint of the head and torso, and the position of the eyes and nose. Visibility and left/right labels for corresponding joints are also annotated in a person-centric manner. It is possible to access the dataset at: http://human-pose.mpi-inf.mpg.de/

Common objects in context (COCO) dataset (Lin et al., 2014). The COCO dataset, initially designed for everyday object detection and segmentation in natural settings, has evolved into a large-scale resource. Its applications extend beyond its original scope to include image captioning and keypoint detection. The dataset's images are sourced from Google, Bing, and Flickr image searches, featuring isolated or pairwise object categories. Annotations are carried out via Amazon Mechanical Turk. The complete dataset comprises over 200,000 images, with 250,000 labeled person instances. To specifically cater to HPE, two distinct datasets, namely COCO keypoints 2016 and COCO keypoints 2017, have been curated, corresponding to public keypoint detection challenges. The only divergence between these versions lies in the train/val/test splitting strategy, allowing for direct cross-year result comparisons. The COCO Keypoint Detection Challenge targets the localization of people's keypoints in uncontrolled images. Annotations for each person include 17 body joints, along with visibility and left/right labels, as well as instance human body segmentation. It is possible to access the dataset at: https://cocodataset.org/

Frames labeled in cinema (FLIC) dataset(Sapp and Taskar, 2013). It comprises 5003 images sourced from popular Hollywood movies. To create this dataset, a person detector was applied to every tenth frame of 30 movies, resulting in around 20,000 person candidates. These candidates were then submitted to Amazon Mechanical Turk for obtaining ground truth labeling for 10 upper body joints. Following this, images featuring person occlusion or severely non-frontal views were manually removed. The initial, undeleted set, known as FLIC-full, includes occluded, non-frontal, or mislabeled examples (20,928 in total). Additionally, in the work by Tompson et al. (2014), the FLIC-full dataset undergoes further refinement to create FLIC-plus, ensuring that the training subset does not contain any images from the same scenes as the test subset. It is possible to access the dataset at: https://bensapp.github.io/flic-dataset.html

AI challenger human keypoint detection (AIC-HKD) dataset (Wu et al., 2017). It boasts the most extensive collection of training examples, comprising 210,000, 30,000, 30,000, and 30,000 images designated for training, validation, test A, and test B, respectively. These images, depicting various aspects of people's daily lives, were sourced from Internet search engines. Following the removal of inappropriate examples as well as images with overly small or crowded human figures, everyone in the images was meticulously annotated with a bounding box and 14 keypoints. Each keypoint includes information about visibility and left/right labels. Beyond the static image datasets, the AIC-HKD dataset also incorporates densely annotated video frames, bringing it closer to real-life application scenarios. These video datasets offer the potential to leverage temporal information and are suitable for tasks like action recognition. The dataset is not publicly available anymore.

Table 4
Dataset comparisons among the 2D datasets.

Dataset name	#Videos	#Images	#Activities	#Persons	#Joints	Multi-Person
LSP Dataset (Johnson and Everingham, 2010)	–	2,000	8	2,000	14	No
LSP-extended Dataset (Johnson and Everingham, 2010)	–	10,000	3	10,000	14	No
Penn action Dataset (Zhang et al., 2013)	2,326	159,633	15	–	13	No
JHMDB Dataset (Jhuang et al., 2013)	5,100	31,838	21	–	15	No
sub-JHMDB Dataset (Jhuang et al., 2013)	316	–	12	–	15	No
MPII Human Pose Dataset (Andriluka et al., 2014)	–	24,920	410	–	16	Yes
COCO Dataset (Lin et al., 2014)	–	>200,000	–	>250,000	17	Yes
FLIC Dataset (Sapp and Taskar, 2013)	–	5,003	–	–	10	Yes
FLIC-full Dataset (Sapp and Taskar, 2013)	–	20,928	–	~20,000	10	Yes
AIC-HKD Dataset (Wu et al., 2017)	–	300,000	–	–	14	Yes
CrowdedPose (Li et al., 2019b)	–	20,000	–	~80,000	14	Yes
PoseTrack17 Dataset (Andriluka et al., 2018)	550	~23,000	–	153,615	15	Yes
PoseTrack18 Dataset (Andriluka et al., 2018)	763	46,933	–	>200,000	15	Yes
PoseTrack21 Dataset (Doering et al., 2022)	763	46,933	–	>420,000	15	Yes
HiEve dataset (Lin et al., 2023)	32	49,820	>56,000	–	14	Yes

CrowdedPose dataset (Li et al., 2019b). It has been proposed to improve model performance in crowded scenarios and enhance generalization across different situations. It comprises 20,000 images containing approximately 80,000 individuals. The dataset is collected by analyzing three public benchmarks, dividing their images into 20 groups based on the Crowd Index ranging from 0 to 1, with a step size of 0.05. Then, 30,000 images are uniformly sampled from these groups. It is possible to access the dataset at: https://github.com/Jeff-sjtu/CrowdPose

PoseTrack dataset (Andriluka et al., 2018). It is a dataset for HPE, and articulated tracking is notably extensive and diverse in terms of data variability and complexity. It features annotations for 15 body parts in each body pose, including head, nose, neck, shoulders, elbows, wrists, hips, knees, and ankles. It was originally released in two versions. The PoseTrack 2017 dataset follows the split of the MPII Human Pose dataset, which splits the dataset into 292, 50, and 208 videos for training, validation, and testing. In total, PoseTrack 2017 provides around 23,000 labeled frames with 153,615 annotated poses, where each pose is annotated with 15 keypoints. PoseTrack 2018 extends the previous dataset and contains 593, 170, and 375 videos for training, validation, and testing. In total, PoseTrack 2018 consists of 46,933 labeled frames for the training and validation sets because the test set is not publicly available (https://posetrack.net). Unfortunately, its original release is no longer public, but an updated version has been recently released, PoseTrack21 (Doering et al., 2022). The dataset has been completely re-annotated, and more person instances, also smaller ones, have been properly identified and annotated. More improvements and fixes over the training and validation sets of PoseTrack 2018 have been performed to make it more robust and reliable. It is possible to access the dataset at: https://github.com/andoer/PoseTrack21

Human-centric video analysis in complex events (HiEve) dataset(Lin et al., 2023). The dataset comprises videos from two sources: those obtained by the researchers with participant consent, and others collected from online repositories like YouTube. These videos feature from 12 different scenes: airport, dining hall, factory, lounge, stadium, jail, mall, square, school, station and street. With a total of 32 video sequences, most are longer than 900 frames, capturing real-world scenarios with complex events. The dataset contains 49,820 frames, featuring 1,099,357 poses, 56,643 action instances, with an average length of 485 frames. Events depicted in the videos include fighting, quarreling, accident, robbery, after-school, shopping, getting-off, dining, walking, playing and waiting. Annotations were manually performed by a professional annotation company with experienced annotators, known for their work on various benchmarks. It is possible to access the dataset here: http://humaninevents.org/

5.2. Popular datasets for 3D HPE

5.2.1. Multi-person dataset

Current 3D pose datasets often exhibit biases towards specific environments like indoor settings and feature-limited actions. For instance, the prominent Human3.6M dataset includes only 11 actors performing 15 activities. In contrast, 2D pose datasets (Andriluka et al., 2014; Lin et al., 2014; Johnson and Everingham, 2010) offer richer poses and environments and are consequently frequently employed to enhance the generalization of 3D algorithms. Many existing 3D pose estimation methods utilize 2D pose as an intermediate representation (Zhao et al., 2019; Sun et al., 2019b) or even as network input (Martinez et al., 2017; Pavllo et al., 2019) to alleviate complexities. However, compared to 2D pose datasets, 3D pose ones lack diversity in poses and environments. To address this issue, various approaches aim to train models in an unsupervised or weak-supervised manner. For example, some methods utilize weak but cost-effective labels for supervision. For instance, Pavlakos et al. (2018a) leverage weak ordinal depth relations between keypoints for supervision, achieving comparable performance to direct supervision using ground truth 3D pose annotations. Similarly, Zhou et al. (2019) encode explicit depth ordering of adjacent keypoints through a heatmap triplet loss as ground truth. Unlike HumanEva-I, this later dataset contains a relatively small test set of synchronized frames (~2500). A summary of relevant statistics about 3D image datasets is reported in Table 5.

5.2.2. Single-person dataset

HumanEva-I&II datasets (Sigal et al., 2010). HumanEva-I was captured earlier and is the larger set of the two. It features data from four subjects performing six predefined actions in three repetitions, captured in both video and motion capture formats. HumanEva-II, on the other hand, was captured using a more advanced hardware system, resulting in higher-quality motion capture data and improved hardware synchronization. This set includes data from two subjects performing an extended sequence of actions known as "Combo". Both datasets were captured in a laboratory setting without any external occlusions or significant clutter. The annotations for both sets were obtained using a commercial Motion Capture (MoCap) system provided by ViconPeak. HumanEva-I comprises seven-view video sequences, consisting of four grayscale and three color views, precisely synchronized with corresponding 3D body poses. The capture area measures 3 m × 2 m, and the actions performed include walking, jogging, gesturing, throwing and catching a ball, boxing, and the Combo action. HumanEva-II, an extension of HumanEva-I, focuses exclusively on the Combo action and involves only two subjects. Both subjects from HumanEva-II also appear in the HumanEva-I dataset. The dataset is available at: http://humaneva.is.tue.mpg.de/

Table 5
Dataset comparisons among the 3D datasets.

Dataset name	#Videos	#Images	#Activities	#Persons	Data source	#Joints	Multi-Person
HumanEva-I (Sigal et al., 2010)	56	~80,000	6	4	6-MoCap 3-RGB 4-BW	15	No
HumanEva-II (Sigal et al., 2010)	–	2460	1/Combo	2	12-MoCap 4-RGB	15	No
Human3.6M Dataset (Ionescu et al., 2014)	–	~3.578M	17	11	10-MoCap 4-RGB 1-Time-of-flight sensor	32	No
MPI-INF-3DHP (Mehta et al., 2017a)	–	>1.3M	8	8	14-RGB	17	No
MoVi (Ghorbani et al., 2020)	1054	–	20	90	15-MoCap 2-SyncCam 2-RGB 18-IMUs	20	No
SURREAL Dataset (Varol et al., 2017)	~67K	~6.536M	23	145	Syntetic Generated	24	No
AMASS (Mahmood et al., 2019)	–	11,265	–	344	Merge of Motion Capture Datasets	52	No
UP-3D (Lassner et al., 2017)	–	6,777	–	5,569	LSP datasets + MPII-HumanPose dataset	–	No
Monocular TotalCapture (Xiang et al., 2019)	–	834K body +111K hands	1/Combo	40	31 multi view cameras	–	No
3DPW (von Marcard et al., 2018)	60	>51,000	–	7	1-RGB 17-IMUs/9-IMUs	18	Yes
CMU Panoptic (Joo et al., 2015)	–	~297K frames ~154M images	4 Games	8	480-SyncCam 31-HDCams 10-Kinects	15	Yes
JTA Dataset (Fabbri et al., 2018)	512	~460,000	–	–	Photorealistic VideoGame	22	Yes
DensePose (Lassner et al., 2017)	–	50K	–	–	COCO Dataset	–	Yes
NTU RGB+D (Liu et al., 2020)	>114k	>8M	120	106	3 Kinect V2	25	Yes

Human3.6M dataset (Ionescu et al., 2014). It stands as a pivotal resource in the domain of HPE, offering a comprehensive collection of annotated human activities captured in controlled environments. Developed by the Institute of Robotics and Intelligent Systems at ETH Zurich, the dataset is renowned for its high-quality annotations and diverse range of human poses, making it a foundational benchmark for evaluating pose estimation algorithms. The dataset is gathered from an environment equipped with 15 sensors, including digital video cameras, a time-of-flight sensor, and motion cameras. The designated laboratory area is approximately 6 m x 5 m, cce of about 4 m x 3 m where subjects are fully visible in all video cameras. The motions were performed by 11 professional actors across 17 scenarios, including actions like walking with asymmetries, sitting, lying down, and various waiting poses, resulting in 3.6 million different human poses from four different views. Each pose is gathered using a precise marker-based made of 10 Motion Capture (MoCap) cameras, 4 RGB cameras, and 1 Time-of-flight sensor. The availability of ground truth annotations in the Human3.6M dataset makes it a valuable resource for benchmarking algorithm performance and advancing the state-of-the-art in HPE research. It is available at: http://vision.imar.ro/human3.6m. For the evaluation, three different protocols are mostly used. Protocol #1 refers to the evaluation in which are used 5 subjects (S1, S5, S6, S7, S8) for training and the rest 2 subjects (S9, S11) for evaluation (Martinez et al., 2017; Pavlakos et al., 2017a) with the metric MPJPE. In Protocol #2, the prediction has been further aligned with the ground truth via a rigid transformation, i.e. translation, rotation, and scale (PA-MPJPE) (Li et al., 2023; Martinez et al., 2017; Hossain and Little, 2018; Bogo et al., 2016). Protocol #3 refers to the evaluation in which the training is made with six subjects (S1, S5, S6, S7, S8 and S9), and every 64th frame of the frontal view of S11 is used for testing (Rogez and Schmid, 2016; Yasin et al., 2016).

MPI-INF-3DHP (Mehta et al., 2017a). The dataset comprises real humans with ground truth 3D annotations from a markerless motion capture system. It offers diverse characteristics, including everyday clothing appearance, various motions, interactions with objects, and varied camera viewpoints. Recorded in a multi-camera studio with green screen technology, it enables automatic segmentation and augmentation. Featuring 8 actors performing 8 activity sets each, including walking, sitting, exercise poses, and dynamic actions, with each activity set spanning roughly one minute. Actors wear two sets of clothing for versatility in augmentation. The dataset includes a wide range of viewpoints captured by 14 RGB cameras, totaling over 1.3 million frames, with approximately 500k frames from five chest-high cameras. Both true 3D annotations and a skeleton are provided. It is possible to access the dataset at: https://vcai.mpi-inf.mpg.de/3dhp-dataset/

Motion and video (MoVi) dataset (Ghorbani et al., 2020). It is a large-scale single-person video dataset featuring joint angles and joint 3D locations along with kinematic trees, occlusions, and optical marker data. It includes 90 actors performing 20 predefined everyday actions and sports movements, as well as one self-chosen movement. The dataset consists of recordings from five capture rounds using 15 MoCap cameras, 2 Synchronized RGB cameras (SyncCam), 2 Standard RGB cameras, and 18 inertial measurement units (IMUs). IMUs consist of multiple sensors, such as accelerometers, gyroscopes, and magnetometers, that work in tandem to provide comprehensive motion data. Actors were recorded wearing natural clothing in some rounds and minimal clothing in others. Overall, the dataset comprises 9 h of motion capture data, 17 h of video data from 4 different viewpoints, and 6.6 h of IMUs data. It is possible to access the dataset at: https://www.biomotionlab.ca/movi/

SURREAL dataset(Varol et al., 2017). It is distinguished by its synthetic nature, providing a unique and diverse collection of photorealistic human images and annotations generated through sophisticated computer graphics techniques. This dataset offers a novel approach to data generation, enabling the creation of highly realistic human models in varying poses, clothing, and environments, thereby addressing limitations associated with traditional datasets that rely on real-world data collection. It includes over 6 million frames with ground truth pose, depth maps, and segmentation masks, providing rich pixel-wise ground truth information. The dataset comprises 145 subjects, 2607 sequences, 67,582 clips, and 6,536,752 frames in total. Each human body is generated with a random 3D pose, shape, and texture, rendered from a random viewpoint for some random lighting and a random background image. The random samples are taken from a common dataset available in the literature. As an example, for the Human texture, 20% of data is rendered with the first set (158 CAESAR textures (Rogez and Schmid, 2016) randomly sampled from 4000), and the rest with the second set (772 clothed textures). The dataset is available at: https://www.di.ens.fr/willow/research/surreal/data/

AMASS (Mahmood et al., 2019). It is a comprehensive database of human motions that integrates various optical marker-based motion capture datasets into a unified one. One of the distinguishing features of the AMASS dataset is its representation of human motion as temporally coherent surface deformations, allowing for the analysis of both motion dynamics and body shape variations. With detailed annotations and a standardized data structure, the dataset facilitates cross-dataset comparisons, algorithm benchmarking, and the development of novel techniques for motion analysis and synthesis. It is valuable for animation, visualization, and generating training data for deep learning. The dataset contains over 40 h of motion data, involving more than 300 subjects and encompassing over 11000 frames. The dataset is available at: https://amass.is.tue.mpg.de/

Unite the people: UP-3D (Lassner et al., 2017). It is a dataset that "Unites the People" of different datasets for multiple tasks. This dataset combines two LSP datasets and the single-person part of the MPII-HumanPose dataset. It includes high-fidelity semantic body part segmentation in 31 parts and 91 landmark human pose estimations. It consists of 5569 training images and 1208 test images with 5569 subjects. It has been human annotated for reliability, with an interactive annotation tool on top of the Opensurfaces package to work with Amazon Mechanical Turk. The interactive Grabcut algorithm has been used to obtain image-consistent silhouette borders. There are several types of annotations for the same images, and a clear nomenclature is important: the datasets with the prefix "UP" (for Unite the People, optionally with an "i" for initial, i.e., not including the FashionPose dataset). Followed by a dash, they specify the type of the annotation (Segmentation, Pose, or 3D) and the granularity. If the annotations have been acquired by humans, it ends with "h". The annotations are freely available for academic and non-commercial use. About the images, the original dataset license must always be used. They only provide our cutouts for convenience and reproducibility. The datasets are linked from their respective thumbnail image. Each image in UP-3D is also provided with a file including quality information ('medium' or 'high') of the 3D fits. 'Medium' means that the rough areas of the body overlap with the image evidence. Rotation of the limbs is not necessarily correct. A 'high' rating indicates that limb rotation is also mostly correct. The pose datasets use only images with a 'high' rating. The dataset is available at: https://files.is.tuebingen.mpg.de/classner/up/#datasets

Monocular total capture dataset (Xiang et al., 2019). It is a subset of our Panoptic Studio Dataset, recorded with 31 HD cameras, with 40 subjects, where each one had to follow the motion in the same video proposed of around 2.5 min recorded in advance. This has been processed with multi-view 3D reconstruction algorithms to reconstruct 3D body, hand, and face keypoints. By reusing the results of these algorithms, the average lengths of all bones for every subject have been computed, and frames with the difference between the length of any bone in the frame and the average length above a certain threshold have been discarded. Manually, the correctness of hand annotations by projecting the skeletons onto three camera views has been verified, and the alignment between the projection and images has been checked. The result is a total of 834K image-annotation pairs for bodies and 111K pairs for hands. The dataset is available at: http://domedb.perception.cs.cmu.edu/mtc.html

3D poses in the wild (3DPW) dataset(von Marcard et al., 2018). It is the first of its kind to provide accurate 3D poses for evaluation in real-world scenarios. Each annotated pose in the dataset is represented in 3D space, allowing researchers to analyze the spatial configuration of human body joints accurately. The 3DPW dataset's emphasis on real-world scenarios and uncontrolled settings makes it a valuable resource for assessing the robustness and generalization capabilities of pose estimation models across different contexts. Unlike other outdoor datasets, which are limited in recording volume, 3DPW features video clips are captured from a moving phone camera. It consists of 60 video sequences, more than 51,000 frames, showcasing various activities such as climbing, golfing, relaxing on the beach, etc., along with 2D and 3D pose annotations, camera poses for each frame, and eighteen 3D models in different clothing variations. For single-subject tracking, they attached 17 IMUs to all major bone segments. Then 9-10 IMUs per person are used to simultaneously track up to 2 subjects. The dataset is available at: https://virtualhumans.mpi-inf.mpg.de/3DPW/license.html

CMU panoptic (Joo et al., 2015). It is a dataset that captures people engaged in various social interactions using a camera system with 480 views obtained from 480 synchronized cameras, 31 RGB HD cameras, and 10 Kinects. Participants are engaged in various games, including Ultimatum, Prisoner's Dilemma, Mafia, Haggling, and 007-bang game. These games were chosen to study conflict and cooperation, and induce natural interactions. Sessions varied in participants from three to eight. From the captured data, five vignettes with interesting non-verbal interactions were selected. The dataset, including 480 synchronized camera feeds calibration, 3D trajectory stream, 3D pose reconstruction, and articulated nonrigidity representation result, will be publicly shared. It contains 60 h of data with 3D poses and tracking information. The dataset is available at: http://domedb.perception.cs.cmu.edu/

Joint track auto (JTA) dataset (Fabbri et al., 2018). It offers extensive data for pedestrian pose estimation and tracking in urban settings, utilizing the highly detailed video game *Grand Theft Auto V*. It includes numerous body poses in diverse urban environments with varying lighting and perspectives. The videos feature up to 60 people per scene on average, totaling nearly 10 million annotated body poses across densely annotated frames. The dataset comprises 512 Full HD videos, each 30 s long, recorded at 30 fps. Automatic annotation of 14 body parts is facilitated through game editing, and precise 3D information is provided for each annotated joint, including coordinates in both 2D image space and 3D simulation space. Unlike other datasets, the absolute scale for each pedestrian is determined through 3D annotation rather than head bounding boxes. The dataset is available at: http://imagelab.ing.unimore.it/jta

DensePose dataset(Güler et al., 2018). It is a large-scale ground-truth dataset with image-to-surface correspondences manually annotated on 50K COCO images. To build this dataset Facebook AI Research team involved human annotators, who were establishing dense correspondences from 2D images to surface-based representations of the human body using a specifically developed annotation pipeline. In the first stage, annotators define regions corresponding to visible, semantically defined body parts. In the second stage, every part region is sampled with a set of roughly equidistant points, and annotators are requested to bring these points in correspondence with the surface. The researchers wanted to avoid manual rotation of the surface, and for this purpose, they provided annotators with six pre-rendered views of the same body part and allowed users to place landmarks on any of them. DensePose is the first manually collected ground truth dataset for the task of dense human pose estimation. The dataset is available at: http://densepose.org/

NTU RGB+D dataset (Liu et al., 2020). Even though it is not widely used for HPE, the NTU RGB+D 120 dataset, an extension of the NTU RGB+D dataset (Shahroudy et al., 2016), is a video collection captured with RGB+D cameras, designed specifically for human action recognition. The *120* in the dataset's name represents its 120 action classes, encompassing a total of 114,480 video samples. This dataset includes various data types: RGB videos, depth map sequences, 3D skeletal data with 25 body joints per frame, and infrared videos for each sample, all recorded simultaneously by three Kinect V2 cameras. Multi-person interactions, with up to two people, cover actions such as hugging, pushing, and passing objects to another person. Access to the dataset can be requested via the following link: https://rose1.ntu.edu.sg/dataset/actionRecognition/

6. Evaluation and metrics

Analyzing performance of HPE systems is crucial to determining their efficacy, resilience, and capacity of generalization over a variety of datasets and real-world situations. The main metrics, procedures, and assessment techniques used in the evaluation of HPE systems, both 2D and 3D, are covered in this section. This provides insights into the standards used to gauge the accuracy, precision, completeness, and other performance indicators that establish the effectiveness and dependability of pose estimation algorithms and models.

6.1. Evaluation metrics for 2D HPE

Assessing the accuracy of HPE proves challenging due to numerous factors and criteria to account for, such as the extent of coverage, whether it is single or multiple pose estimation, and the scale of the human body. Consequently, a variety of evaluation metrics have been employed for 2D HPE. In this context, we provide an overview of the frequently utilized metrics.

Percentage of correct parts (PCP) (Eichner et al., 2012). It is a common metric in early 2D HPE research, assessing the accuracy of limb localization by comparing predicted and ground truth joint distances within a fraction of limb length (typically 0.1 to 0.5). While widely used in single-person HPE evaluation, it is less prevalent in recent research due to its bias towards penalizing shorter limbs. Formally, PCP (Eq. (1)) is the ratio between the *"Number of Correct Parts"* that refers to the count of correctly detected limbs (e.g., arms, legs) based on the predicted joint locations and the *"Total Number of Parts"* represents the total number of limbs in the dataset.

$$\mathrm{PCP} = \frac{\text{Number of Correct Parts}}{\text{Total Number of Parts}} \tag{1}$$

Percentage of detected joints (PDJ) (Toshev and Szegedy, 2014). This metric is introduced to address this issue of using a scale factor related to the limb length in PCP. Indeed, PDJ considers a predicted joint as detected if its Euclidean distance (Eq. (2)) from the true joint is within a fraction of the torso diameter. By incorporating the torso diameter as a reference point for joint proximity evaluation, PDJ enhances the precision and granularity of performance evaluation, providing deeper insights into the efficacy and robustness of pose estimation algorithms in capturing intricate human poses with greater fidelity and accuracy. Given the ground truth joint location j and the predicted joint location p, the Euclidean distance d is calculated as:

$$d = \sqrt{(x_j - x_p)^2 + (y_j - y_p)^2} \tag{2}$$

PCPm (Andriluka et al., 2014). In accordance with the ideas of PCP, the modified version known as PCPm uses a similar algorithmic framework to assess the accuracy of HPE systems. One important difference is that, in PCPm, the threshold for matching predicted and ground-truth joints is set at 50% of the mean ground-truth segment length. By improving the sensitivity and specificity of joint matching criteria, this customized method seeks to provide a more comprehensive evaluation of pose estimate performance in precisely capturing anatomical landmarks and articulations. However, it is crucial to remember that PCP metrics, such as PCPm, might be subject to foreshortening effects, in which changes in perspective and distance can introduce distortions in body part measurements, potentially impacting the overall precision and reliability of pose estimation evaluations across diverse viewing angles and spatial configurations.

Percentage of correct keypoints (PCK) (Yang and Ramanan, 2013). PCK measures if the predicted keypoint is in a certain threshold with respect to the true joint (PCK@threshold). This threshold can be defined in different ways, such as the percentage of the length connecting two joints (like the head bone), a fraction of the subject's width (like the torso diameter), or a fixed value like 150 mm. E.g. PCK@0.2 is the PCK with distance threshold used for evaluation of 0.2 of torso diameter, i.e., the distance between the predicted and true joint must be less than 0.2 times the torso diameter.

Average precision (AP), mean average precision (mAP) and average recall (AR) (Yang and Ramanan, 2013). Average Precision takes into account both the accuracy (i.e., the number of correctly recognized keypoints) and the completeness (i.e., the accuracy of each keypoint's position) of the model's predictions. It determines the average accuracy at various thresholds of "correct" keypoint positions. In order to identify if a keypoint is "correct" the Object Keypoint Similarity (OKS) is commonly used. It assesses pose estimation accuracy in real-world settings without the need for pre-defined bounding boxes around individuals, and it provides a standardized way to compare the predicted keypoints with the ground truth keypoints. It normalizes the distance between predicted points and ground truth points by the scale of the object s defined as the square root of the object segment area. In the COCO dataset, AP is calculated as the average of AP with OKS=0.50:0.05:0.95. (i.e., medium, loose, and strict similarity), following Eq. (3).

$$\mathrm{AP} = \frac{\sum_r P@r}{R} \tag{3}$$

$P@r$ represents precision at the r_{th} OKS threshold. R is the total number of relevant keypoints at the r_{th} OKS threshold. For multi-person pose estimation, Mean Average Precision (mAP) is used. Indeed, it is the average of AP for each person in the frame. Similar to AP, AR calculates the recall across various thresholds for what constitutes a "correct" keypoint location (closeness to the ground truth location). Recall represents the proportion of true positive keypoints that the model correctly identified. In simpler terms, it tells how well the model captures all the keypoints in the image.

6.2. Evaluation metrics for 3D HPE

Mean per joint position error (MPJPE). MPJPE is the most widely used metric to evaluate the performance of 3D HPE (Wu and Xiao, 2020). MPJPE (Eq. (4)) is calculated as the mean Euclidean distance between the predicted 3D joint locations and the corresponding ground truth joint locations. The formula for MPJPE is expressed as follows:

$$\mathrm{MPJPE} = \frac{1}{N}\sum_{i=1}^{N} \|J_i - J_i^*\|_2 \tag{4}$$

where N represents the number of joints, J_i and J_i^* denote the ground truth position and the estimated position, respectively, of the i_{th} joint.

Procrustes aligned mean per joint position error (PA-MPJPE). PA-MPJPE (Arnab et al., 2019) extends MPJPE evaluation by aligning the estimated pose with the ground truth through Procrustes Analysis, with translation, rotation, and scale (Luo and Hancock, 1999). This statistical shape analysis method ensures a rigid alignment between the predicted and ground truth poses, removing translation, rotation, and scale effects.

Normalized mean per joint position error (NMPJPE). NMPJPE metric aims to normalize the error by the scale of the human body or by the inter-joint distances in the ground truth poses. This normalization helps to provide a more relative measure of pose estimation accuracy, especially when dealing with poses of different scales or proportions. The formula for NMPJPE is expressed as follows:

$$\mathrm{NMPJPE} = \frac{1}{N}\sum_{i=1}^{N} \frac{\|J_i - J_i^*\|_2}{S_i} \tag{5}$$

where S_i represents the normalization factor for the i_{th} joint, which could be the scale of the human body or the inter-joint distance.

Mean per vertex error (MPVE). MPVE measures the average Euclidean distance between each vertex and its corresponding ground truth location:

$$\mathrm{MPVE} = \frac{1}{N}\sum_{i=1}^{N} \|V_i - V_i^*\|_2 \tag{6}$$

where N represents the total number of vertices, V_i and V_i^* denote the position of the i_{th} vertex in the reconstructed surface and its corresponding ground truth position, respectively.

3D percentage of correct keypoints (3DPCK). It is essentially the 3D counterpart of 2D Percentage of Correct Keypoints (PCK). It quantifies the percentage of joints accurately estimated within a specified threshold distance from their ground truth positions. This threshold distance is usually fixed to 150 mm or determined as a fraction of the bounding volume encompassing the human body or as a fraction of the average inter-joint distance.

The mean per joint angle error (MPJAE). Similar to the MPJPE, which measures the Euclidean distance between the predicted and ground truth joint positions, the MPJAE focuses on the angular difference between the estimated joint angles and the ground truth joint angles. MPJAE is computed by measuring the absolute difference between the joint angles of the estimated pose and those of the ground truth pose. This calculation is performed for each joint that possesses one degree of freedom. For instance, in the case of the elbow joint, the angle between the upper arm and the lower arm is measured. For joints with three degrees of freedom, such as the shoulder joint often represented using ZXY Euler angles, the angle difference is calculated separately for each degree of freedom. This means that the angle differences for rotation around the Z-axis, rotation around the X-axis, and rotation around the Y-axis are computed individually. The final value is averaged based on the degree of freedom.

Procrustes aligned mean per joint angle error (PA-MPJAE). PA-MPJAE represents an improved iteration of MPJAE, integrating Procrustes Analysis to refine its accuracy assessment. Procrustes Analysis, a statistical method, harmonizes two sets of geometric shapes by optimizing the translation, rotation, and scaling of one set to match the other closely. In the realm of PA-MPJAE, this technique, known as Procrustes Alignment, aligns the estimated pose with the ground truth before analyzing joint angle errors. To compute PA-MPJAE, the estimated pose undergoes alignment with the ground truth via Procrustes Analysis. Next, absolute differences in joint angles between the aligned estimated pose and the ground truth are computed for each joint. The mean of these differences across all joints returns the PA-MPJAE value. This alignment procedure effectively minimizes discrepancies in translation, rotation, or scale between the estimated and ground truth poses, refining the accuracy assessment in PA-MPJAE.

6.3. Models performances comparison

In this section, we present a comprehensive evaluation of the performance of several state-of-the-art HPE models, all of which have been trained and tested on the challenging COCO, MPII, LSP, PoseTrack17, Human3.6M, MPI-INF-3DHP datasets. By comparing the results of these models, we aim to provide a deeper understanding of the trade-offs between different design choices, such as the number of parameters, the complexity of the network, and the choice of the loss function, and to identify the most effective approaches for this challenging computer vision problem. The findings of this comparison will contribute to informing the design of future models that can better address the complexities of this problem.

2D human pose estimation (HPE) methods. Table 6 compares the performance of 2D HPE methods on the COCO (Lin et al., 2014), PoseTrack17 (Andriluka et al., 2018), MPII (Andriluka et al., 2014), and LSP (Johnson and Everingham, 2010) datasets. VitPose (Xu et al., 2022) achieves the highest performance on the COCO dataset. The advantage of VitPose over other architectures is due to the use of the transformer architecture, which is also commonly used in the field of natural language processing and adapted to computer vision tasks such as HPE. Self-Attention Mechanism provides a strong boost in performance, indeed the image is divided into a grid of non-overlapping patches, and each patch is treated as a separate token. The model applies a self-attention mechanism to these patches, allowing it to attend to and weigh the importance of different parts of the image relative to each other. By leveraging the self-attention mechanism and the transformer architecture, VitPose is able to effectively model the complex relationships between different parts of the human body and accurately estimate the 2D locations of the body joints from an image.

For MPII and LSP, the methods proposed in the paper by Bulat et al. (2020) outperform others. They designed a unique architecture based on CNN network, consisting of two interconnected parts, to help the network understand the relationships between different body parts and the surrounding context. This CNN architecture, which combines detection and regression in a cascade, allows the network to (i) focus on the most relevant information, (ii) take into account the constraints of the body's structure, and (iii) make use of the surrounding context to predict the location of the hidden body parts. This strategy makes the model performing much better than other approaches in literature.

Finally, the methodology presented in Liu et al. (2022) with FAMI-Pose demonstrates the best results on PoseTrack17. By making use of the temporal relationships between the frames, DCPose overcomes difficulties in detecting poses when there is motion blur, video defocus, or occlusions. The framework consists of three key components: a Pose Temporal Merger, which uses a type of convolutional layer to combine spatial and temporal information; a Pose Residual Fusion module, which calculates the weighted differences in pose estimates using a basic type of neural network layer; and a Pose Correction Network, which refines the pose estimates using a type of convolutional network called Deformable Convolution V2 (DCN v2). This strategy proved to be successful.

3D human pose estimation (HPE) methods. Three separate tables (Tables 7, 8, and 9) distinguish the performance of 3D HPE methods based on their input data: monocular images, multi-view images, and sequences of monocular images. All tables use the Human3.6M dataset (Ionescu et al., 2014) with both Protocol #1 and Protocol #2, as well as the MPI-INF-3DHP dataset (Mehta et al., 2017a).

- Table 7: For both Human3.6M protocols, the HEMlets (Zhou et al., 2019) model demonstrates the best performance. HEMlets achieves this result by dividing the full body into 14 parts and representing depth information through three types of heatmaps, each indicating the absence, presence, or likelihood of a joint. The approach involves a two-stage process: first, a CNN is trained to predict body parts from input images, and then, the CNN predicts volumetric heatmaps of joints. By integrating the volumetric heatmaps, the network can accurately determine the joint locations, all while learning end-to-end. Notably, their method outperforms existing state-of-the-art methods by around 20% on the Human3.6M dataset.
  On the MPI-INF-3DHP dataset, JoyPose (Du et al., 2024) achieves the best results. It achieves such results thanks to the data augmentation process directed by a reinforcement learning approach. In a similar manner, the anatomy-informed pose feature representation module is designed to capture both global and local features, inspired by the patterns of error observed in pose estimation across various joints of the human body and guided by an understanding of human anatomy and kinematics.
- Table 8: Chun et al. (2023b) and Chen et al. (2019a) achieve the best results on the Human3.6M dataset. For MPI-INF-3DHP, S. Chun et al. (2023a) outperforms other methods. Chun et al. got such great results thanks to the use of CNNs to predict the positions of mesh vertices from input images and then trained the SMPL model on these predicted vertices to obtain the SMPL parameters. The predicted mesh vertices, which contain a wealth of information about joint rotation and shape, are easier to learn compared to the SMPL parameters directly. Moreover, they also showed that incorporating an autoencoder to learn vertex heatmaps significantly enhances the effectiveness of this method. The Vertex Heatmap Autoencoder (VHA) learns the manifold of possible human meshes by capturing the patterns in the data.

Table 6
Accuracy comparison for several state-of-the-art 2D HPE methods and various Datasets. For the COCO and PoseTrack17 datasets the AP metric was used, while for MPII and LSP the PCK@0.5 (except of the frameworks with the * that use PCK@0.2). The best results are shown in **bold**.

Reference	Model	COCO↑	MPII↑	LSP↑	PoseTrack17↑	Single/Multi
Carreira et al. (2016)	–	–	81.3	–	–	Single
Pishchulin et al. (2016)	Deepcut	–	82.4	87.1*	–	Multi
Wei et al. (2016)	CPM	–	88.5	90.5	–	Single
Insafutdinov et al. (2016)	DeeperCut	–	88.5	90.1*	–	Multi
Bulat and Tzimiropoulos (2016)	–	–	89.7	90.7	–	Single
Newell et al. (2016)	Stacked Hourglass	56.6	90.9	–	–	Single
Yang et al. (2017)	–	–	92.0	93.9	–	Single
Cao et al. (2017)	OpenPose	61.8	–	–	–	Multi
Newell et al. (2017)	–	65.5	–	–	–	Multi
Papandreou et al. (2017)	–	68.5	–	–	–	Multi
He et al. (2017)	Mask R-CNN	63.1	–	–	–	Multi
Nibali et al. (2018)	–	–	89.5	–	–	Single
Tang et al. (2018)	PBN	–	92.3	95.1*	–	Single
Papandreou et al. (2018)	PersonLab	68.7	–	–	–	Multi
Kocabas et al. (2018)	MultiPoseNet	69.6	–	–	–	Multi
Chen et al. (2018)	CPN	73.0	–	–	–	Multi
Xiao et al. (2018)	SimpleBaseline	73.7	91.5	–	74.6	Multi
Luvizon et al. (2019)	–	–	91.2	–	–	Single
Tang and Wu (2019)	PBN	–	92.7	94.5	–	Single
Kreiss et al. (2019)	PifPaf	66.7	–	–	–	Multi
Sun et al. (2019a)	HRNet	75.5	92.3	–	74.9	Multi
Bulat et al. (2020)	–	–	**94.1**	**94.8**	–	Single
Luo et al. (2021)	–	70.2	–	–	–	Single
Liu et al. (2021)	DCPose	–	–	–	79.2	Multi
Yang et al. (2021a)	TransPose	75.0	93.5	–	–	Multi
Yuan et al. (2021a)	HRFormer	76.2	–	–	–	Multi
Artacho and Savakis (2021)	OmniPose	79.5	92.3*	–	–	Multi
Liu et al. (2022)	FAMI-Pose	–	–	–	**80.9**	Multi
Xu et al. (2022)	ViTPose	**81.1**	–	–	–	Multi

Table 7
Accuracy comparison for single-person 3D HPE methods from monocular images and various Datasets. For the Human3.6M under Protocol #1 the MPJPE (in mm) is used while, for the Protocol #2 the PA-MPJPE (in mm). For the MPI-INF-3DHP the PCK metric is used. The best results are shown in **bold**.

Reference	Model	Human3.6M - P1↓	Human3.6M - P2↓	MPI-INF-3DHP↑	Methodology
Park et al. (2016)	–	117.3	–	–	2D-3D
Zhou et al. (2016)	–	107.3	–	–	Direct
Bogo et al. (2016)	SMPLify	–	82.3	–	SMPL
Nie et al. (2017)	–	97.5	79.5	–	2D-3D
Tomè et al. (2017)	–	88.4	70.7	–	2D-3D
Lassner et al. (2017)	–	80.7	–	–	SMPL
Mehta et al. (2017a)	3DPoseNet	72.9	54.6	76.5	2D-3D
Pavlakos et al. (2017a)	–	71.9	51.9	–	Direct
Tekin et al. (2017)	–	69.7	50.1	–	2D-3D
Zhou et al. (2017)	–	64.9	–	69.2	2D-3D
Martinez et al. (2017)	–	62.9	47.7	42.5	2D-3D
Sun et al. (2017)	–	59.1	48.3	–	Direct
Omran et al. (2018)	NBF	–	59.9	–	SMPL
Kanazawa et al. (2018)	HMR	88.0	56.8	77.1	SMPL
Yang et al. (2018)	–	58.6	37.7	69.0	2D-3D
Pavlakos et al. (2018a)	–	56.2	41.8	71.9	SMPL
Luvizon et al. (2018)	–	53.2	–	–	Direct
Sun et al. (2018)	Integral	49.6	40.6	–	Direct
Habibie et al. (2019)	–	65.7	49.2	–	2D-3D
Sharma et al. (2019)	–	58.0	40.9	–	2D-3D
Zhao et al. (2019)	SemGCN	57.6	–	–	2D-3D
Wang et al. (2019a)	–	52.6	40.7	71.9	2D-3D
Ci et al. (2019)	LCN	52.7	42.2	74.0	2D-3D
Chen et al. (2019b)	–	51	–	64.3	2D-3D
Wandt and Rosenhahn (2019)	RepNet	89.9	65.1	82.5	2D-3D
Wang et al. (2019b)	NRSfM	83.0	57.5	–	2D-3D
Zhou et al. (2019)	HEMlets	**39.9**	**27.9**	75.3	2D-3D
Li et al. (2020a)	TAG-Net	50.9	30.8	81.2	2D-3D
Zeng et al. (2020)	SRNet	49.9	–	77.6	2D-3D
Luvizon et al. (2021)	–	48.6	–	–	Direct
Xu et al. (2020)	–	49.2	38.9	–	2D-3D
Xu and Takano (2021)	GraphSH	51.9	–	80.1	2D-3D
Choi et al. (2021)	MobileHumanPose	51.4	35.2	–	Direct
Gong et al. (2021)	PoseAug	50.2	39.1	88.9	2D-3D
Zou and Tang (2021)	–	49.4	39.1	86.1	2D-3D
Jiang et al. (2024)	ZeDO	51.4	42.1	–	2D-3D
Du et al. (2024)	JoyPose	47.0	37.0	**94.1**	2D-3D

Table 8
Accuracy comparison for single-person 3D HPE methods from multi-view images and various Datasets. For the Human3.6M under Protocol #1 the MPJPE (in mm) is used while for the Protocol #2 the PA-MPJPE (in mm). For the MPI-INF-3DHP the PCK metric is used. The best results are shown in **bold**.

Reference	Model	Human3.6M - P1↓	Human3.6M - P2↓	MPI-INF-3DHP↑
Pavlakos et al. (2017b)	–	56.9	–	–
Rhodin et al. (2018b)	–	66.8	51.6	–
Tomè et al. (2018)	–	52.8	44.6	–
Rhodin et al. (2018a)	–	131.7	98.2	–
Liang and Lin (2019)	–	79.9	45.1	95.0
Kocabas et al. (2019)	EpipolarPose	51.8	45	–
Chen et al. (2019a)	–	46.3	**41.6**	–
Qiu et al. (2019)	–	26.2	–	–
Iskakov et al. (2019)	–	20.8	–	–
Zhang et al. (2021)	AdaFuse	19.5	–	–
Wandt et al. (2021)	CanonPose	74.3	53.0	–
Reddy et al. (2021)	TesseTrack	18.7	–	–
Chun et al. (2023a)	LMT	17.6	–	**99.4**
Chun et al. (2023b)	–	**15.6**	–	–

Table 9
Accuracy comparison for single-person 3D HPE methods from sequence of monocular images and various Datasets. For the Human3.6M under Protocol #1 the MPJPE (in mm) is used while for the Protocol #2 the PA-MPJPE (in mm). For the MPI-INF-3DHP the PCK metric is used. The best results are shown in **bold**, with the best results that are not solely from a module being <u>underlined</u>.

Reference	Model	Human3.6M - P1↓	Human3.6M - P2↓	MPI-INF-3DHP↑
Du et al. (2016)	–	126.5	–	–
Tekin et al. (2016)	–	125.0	–	–
Tung et al. (2017)	–	–	98.4	–
Mehta et al. (2017b)	Vnect	82.5	–	76.6
Lin et al. (2017)	RPSM	73.1	–	–
Katircioglu et al. (2018)	–	67.3	–	–
Lee et al. (2018)	–	55.8	46.2	–
Dabral et al. (2018)	Temporal PoseNet	52.1	36.3	–
Hossain and Little (2018)	–	51.9	42.0	–
Arnab et al. (2019)	–	77.8	54.3	–
Cai et al. (2019)	–	48.8	39.0	–
Pavllo et al. (2019)	VideoPose3D	46.8	36.5	–
Cheng et al. (2019)	–	42.9	32.8	–
Kundu et al. (2020)	–	50.8	–	–
Yuan et al. (2021b)	SimPoE	56.7	41.6	–
Li et al. (2021)	Lifting Transformer	44.7	36.1	–
Zheng et al. (2021)	PoseFormer	44.3	34.6	88.6
Hu et al. (2021)	–	41.1	32.1	97.9
Hu and Ahuja (2021)	HDVR	47.3	–	–
Li et al. (2022b)	MHFormer	43.0	34.4	93.8
Zhang et al. (2022)	MixSTE	39.8	<u>30.6</u>	94.4
Gholami et al. (2022)	AdaptPose	42.5	34	–
Zhu et al. (2023)	MotionBERT	<u>37.5</u>	32.9	–
Zhao et al. (2023)	PoseFormerV2	45.2	35.6	97.9
Mehraban et al. (2024)	MotionAGFormer	38.4	32.5	98.2
Peng et al. (2024)	KTPFormer	40.1	31.9	<u>**98.9**</u>
Zhang et al. (2024)	APP w. MotionBERT	37.0	31.6	–
Zhang et al. (2024)	APP w. MotionAGFormer-B	**36.2**	**29.5**	**98.9**

The Body Code Predictor (BCP) uses the body, prior learned from VHA, to reconstruct human meshes from multi-view images. BCP's state-of-the-art performance in human mesh reconstruction opens up new avenues for research in representation learning and mesh reconstruction.

- Table 9: The Adaptive Pose Pooling (APP) **module** proposed by Zhang et al. (2024) when combined with MotionAGFormer-B proposed by Mehraban et al. (2024), achieves the best results on the Human3.6M dataset with both Protocol #1 and #2, as well as on the MPI-INF-3DHP dataset. However, this module is primarily implemented as an addition to models that already deliver top performance, such as MotionBERT and MotionAGFormer. If we exclude additional modules, then (Zhu et al., 2023) and Zhang et al. (2022) achieve the best results on Human3.6M with Protocols #1 and #2, respectively. Zhu et al. with MotionBERT obtained outstanding results thanks to the use of a motion encoder trained using a Dual-stream Spatio-temporal Transformer (DSTformer) neural network. This network is capable of capturing long-range spatio-temporal relationships among skeletal joints, which is crucial for tasks like 3D pose estimation and action recognition. After pretraining, the learned motion representations can be adapted to various downstream tasks with minimal additional training. Zhang et al. with MixSTE, similarly, use a Temporal Transformer Block to model the temporal motion of each joint separately and a Spatial Transformer Block to learn the spatial correlation between joints. The overall seq2seq model learns the global coherence between sequences. This means it can improve the accuracy of reconstruction by understanding the entire sequence of movements rather than just individual frames. Finally, Peng et al. (2024) with KTPFormer, followed by Mehraban et al. (2024) demonstrates the best results on the MPI-INF-3DHP dataset. KPTFormer, with its two proposed prior attention modules, Kinematics Prior Attention (KPA) and Trajectory Prior Attention (TPA), aims to leverage the known anatomical structure of the human body and motion trajectory information to enhance the learning of global dependencies and features within multi-head self-attention. MotionAGFormer combines the strengths of transformers and Graph Convolutional Networks (GCNs) to capture both global and local relationships in human motion data.

The framework employs an adaptive fusion mechanism to aggregate features from the transformer and graph streams, allowing it to better learn the underlying 3D structure of human poses. It is relevant to observe that, compared to other top-performing models, such as MotionBERT, MotionAGFormer offers similar performance, though slightly less accurate. However, it is more computationally efficient, especially with its ***B*** variant, which strikes a balance between estimation accuracy and computational cost.

7. Research challenges

Several challenges are still considered partially unresolved in the current scenario of application of HPE mainly discussed in scientific literature.

- **Lack of robustness**: When estimating human poses from an image, humans can locate key points outside the image by referring to visual clues such as the body size. Current technologies do not consider these key points and focus only on the bounded area of a given image (Tang et al., 2023). The lack of robustness in existing estimation methods highlights three critical aspects that we think could be improved: accurate localization of the subject and its parts, accurate extraction of valuable features, and accurate pixel alignment of outputs (Pang et al., 2023).
- **Occlusions**: The abovementioned issue about the robustness of data derived from HPE is closely related to the accuracy of the estimation of the algorithms. Traditional systems used for kinematic assessment, such as those based on conventional cameras, have limitations that hinder their potential application as a gold standard in the field. The ultimate goal of HPE algorithms is to achieve a model with the highest possible accuracy, detecting the three-dimensional coordinates of body key points and identifying poses and actions. Still, several factors affect the precision of these methods. Among these, one of the main issues is *occlusion*, in which one or more joints are obscured from the camera due to numerous variables such as self-occlusion, zooming, blurred objects, and interference by random items. In particular, there are three types of occlusion: self-occlusion, inter-object occlusion, and background occlusion. Self-occlusion occurs when a section of an image overlaps the self, obscuring a part of the image. Inter-object occlusion occurs when the relevant object is obscured by other objects, and background occlusion arises when a background object obscures the monitored objects. Potential directions to mitigate the issue could be the creation of new datasets using data augmentation techniques for generating novel frames, particularly those featuring occluded situations. These augmented data could then be leveraged to improve occlusion robustness by simulating such cases during training. An example approach is to employ a Teacher-Student training model, where the teacher learns the fundamentals of HPE from standard images, and the student learns to acquire the teacher's knowledge from data where parts of the human figure are occluded (Bulat and Tzimiropoulos, 2016; Liu et al., 2021; Hu and Ahuja, 2021; Xu et al., 2018; Mehta et al., 2018; Belagiannis et al., 2014a; Wang et al., 2024; Cheng and Xu, 2024).
- **Crowd detection**: It is a crucial challenge in analyzing algorithms' accuracy because overlaps and occlusions make it much harder to recognize human enclosing boxes and derive pose indications from particular key points. Crowd monitoring is essential as it plays a significant role in many other applications, such as population monitoring, public event management, suspicious activity detection, disaster management, and many more. Even more important is managing algorithms for multi-pose detection, which are still in the initial research phase and have yet to yield significant results. The primary challenge lies in the multitude of pose and scale variations that algorithms must contend with, compounded by the variable presence of individuals and hidden body parts caused by occlusion or truncation and the lack of appropriate and annotated datasets for training and testing the algorithms. One possible research direction, similar to the previous one, is the enrichment of large datasets. Alternatively, given the rise of multi-task models for HPE (Khirodkar et al., 2024), a potential advantage could be gained by those that include various computer vision tasks, such as *Body-Part Segmentation* and *Depth Estimation*, in addition to *Human Pose Estimation* and ***Human Detection***, both independently and in combination (Dong et al., 2019; Cao et al., 2017; Fang et al., 2023).
- **Depth estimation accuracy**: Even if the recent lifting approaches reach good results, estimating depth from two-dimensional images captured by cameras incapable of capturing depth-related coordinates is still an open issue. Despite ongoing research efforts to find models capable of providing as accurate depth estimation as possible, there are still challenges to overcome to ensure a high level of disambiguation. Variations in the two-dimensional camera parameters significantly impact depth accuracy, exacerbating the challenges in predicting depth coordinates when the subject undergoes translation or rotation relative to the camera. Recent approaches to achieving better performance have included temporal information and human spatial information. A possible mitigation action to improve the results further is incorporating the spatial information of objects close to humans. For example, if I had to estimate the depth of a human posture, estimating the depth of nearby objects could provide additional context. While this step may not directly address the next challenge of real-time processing, it could still represent a progressive advancement (Habibie et al., 2019; Wang et al., 2019b; Kocabas et al., 2019; Rhodin et al., 2019).
- **Real-time processing**: Real-time processing in HPE presents a persistent challenge due to the intricate nature of human body dynamics and the high computational load required for accurate joint detection and tracking. Although recent advancements in DL frameworks have significantly enhanced the accuracy of HPE algorithms, achieving the necessary frame rates for applications such as interactive gaming and live video analysis remains elusive. This difficulty is compounded by the need for robust performance across diverse environments and varying body postures, which can strain the limits of current processing capabilities. Consequently, a possible solution is to investigate innovative methods, such as scalable architectures and efficient inference techniques, to bridge the gap between high accuracy and real-time execution, aiming to enable seamless integration of HPE in practical applications (Cao et al., 2017; Bazarevsky et al., 2020; Fang et al., 2023; Wang et al., 2023a).
- **Deployment in resource-constrained environments**: Deployment of HPE systems in resource-constrained environments presents a significant unresolved issue, as these applications often require sophisticated algorithms that demand substantial computational power and memory. In scenarios such as mobile devices, drones, or embedded systems, limited processing capabilities and battery life hinder the implementation of high-performance HPE models that typically run on powerful GPUs. While efforts have been made to create lightweight models that maintain accuracy, challenges persist in balancing model complexity with efficiency. Additionally, these environments may lack reliable internet connectivity, complicating the use of cloud-based solutions for processing. Consequently, a possible solution is to investigate more efficient algorithms and compression techniques to enable effective pose estimation in real-time on devices with constrained resources, ensuring that such systems can be widely adopted in various practical applications, from healthcare monitoring to sports analytics (Choi et al., 2021; Yu et al., 2024).

8. Conclusion

This review summarizes the recent progress in HPE from RGB images and videos, along with a brief overview of Pose Estimation Using Non-Camera Technologies. The article highlights influential models for human pose estimation and categorizes tasks such as 2D HPE from images, 2D HPE from videos, 3D single-person PE, and 3D multi-person PE. For the 3D methods, it discusses approaches for Single monocular images, Multi-view images, Sequence of monocular images, and Sequence of multi-view images.

The work for this manuscript began in early 2024 with the selection of articles carried out gradually, collecting and investigating the most cited papers published in journals or high-impact conference proceedings from 2012 until 2024. The inclusion criterion was the number of citations. In order to be fair and include also the newest publications, different thresholds for the number of citations were applied, depending on the date of the publication: for papers published before 2021, the threshold was set at 50 citations, while for those published afterward, the threshold was set at 15 citations. For recent papers where citation counts were unreliable, we evaluated them based on quality, novelty, and the impact of their experimental results. The final selection of papers for our initial submission was completed in early June 2024, with a few additional works included on October 2024, during the manuscript revision.

The article begins by introducing HPE applications and related open issues. It also provides a summary of datasets and metrics for HPE. We aim to inspire readers with valuable information while motivating new research efforts to advance the field concerning estimation speed acceleration, unbalanced data augmentation based on unlabeled data sets, as well as occlusion-resistant pose estimation.

As the field continues to evolve, it is essential to acknowledge the challenges that remain, including the need for more diverse and large-scale datasets, as well as the requirement for more robust and generalizable methods that can effectively handle the complexities of real-world scenarios. As researchers, we are excited to build upon the foundations laid by the latest research and to continue pushing the boundaries of what is possible in human pose estimation, ultimately driving the development of more accurate, efficient, and practical applications in a wide range of fields, from healthcare to entertainment and beyond.

CRediT authorship contribution statement

Gaetano Dibenedetto: Conceptualization, Writing - Original Draft. **Stefanos Sotiropoulos:** Conceptualization, Writing - Original Draft. **Marco Polignano:** Conceptualization, Supervision, Writing - Review and Editing. **Giuseppe Cavallo:** Project administration. **Pasquale Lops:** Supervision, Writing - Review and Editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Gaetano Dibenedetto reports financial support was provided by European Union. Marco Polignano reports financial support was provided by European Union. Pasquale Lops reports financial support was provided by European Union. Stefanos Sotiropoulos reports financial support was provided by Puglia Region. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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